Polynomial Commitments for Galois Rings and Applications to SNARKs Over $\mathbb{Z}_{2^k}$

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Abstract. Succinct non-interactive arguments of knowledge (SNARKs) allow a weak verifier to delegate computation tasks to a powerful prover in a verifiable way. However, most SNARK constructions require the computation tasks to be represented as arithmetic circuits over finite fields, incurring a significant overhead when applied for delegations of modern computer programs, which typically are of $\mathbb{Z}_{2^{64}}$ or $\mathbb{Z}_{2^{32}}$ arithmetics. The only exception is Rinocchio (JoC 2023), which builds the first SNARK for rings and features constant proof size. Due to $\mathbb{Z}_{2^k}$ being lack of large evaluation sets for polynomial interpolations, Rinocchio resorts to Galois ring extensions of $\mathbb{Z}_{2^k}$. However, Rinocchio is designated-verifier and the concrete efficiency is unclear at the moment. Towards building *publicly verifiable* SNARKs for rings $\mathbb{Z}_{2^k}$, we follow the well-established framework of polynomial interactive oracle proofs (IOP)-based SNARKs, and obtain the following results:

i) We systematically study the proximity gaps for Reed-Solomon codes over Galois rings. We prove that for Galois rings, the state-of-the-art $(1-\rho)/2$ gap for finite fields given by Ben-Sasson et al. (JACM 2023) still holds. This allows to construct efficient polynomial commitment schemes for Galois rings based on Reed-Solomon codes.
ii) We construct efficient polynomial IOPs for rank one constraint systems (R1CS) over $\mathbb{Z}_{2^k}$ via Galois rings. To amortize the overhead of operating on a large Galois ring extension of $\mathbb{Z}_{2^k}$, we make use of the reverse-multiplication friendly embeddings (RMFEs) techniques.
iii) Combining the above two ingredients together, we put forward the first *publicly verifiable* SNARKs for rings $\mathbb{Z}_{2^k}$ with a transparent setup and plausibly post-quantum security. In addition, we implement and evaluate our constructions. Our evaluations indicate the concrete efficiency is promising, provided that Galois rings operations are optimized well.

1 Introduction

Succinct Non-interactive ARguments of Knowledge (SNARKs) allow a prover to produce a *short* proof of the knowledge of a witness for a given large NP

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statement, such that it can be verified efficiently. If the verification does not require any private information, then it is publicly verifiable, otherwise, it is designated-verifier. Moreover, most SNARKs can be added with the property of *zero-knowledge*, known as zk-SNARKs, meaning that no information is revealed beyond the validity of the statement. Nowadays, SNARKs are very popular and widely used in real-life applications, especially in blockchain.

Typically, modern SNARKs follow a paradigm of combining polynomial interactive oracle proofs (IOP) and polynomial commitments. Specifically, polynomial IOP as an information-theoretic ingredient, is required to be *public-coin*, meaning that the verifier's randomness is public[1], so that Fiat-Shamir heuristic can be applied to make it non-interactive and publicly verifiable. While polynomial commitments are used to instantiate the oracles and rely on cryptographic tools.

Bring SNARKs into Real Life. The past decades have witnessed the rapid development of designing efficient SNARKs, and many [3,13,19,23,32,35,36,48] have been implemented with good concrete efficiency. However, there still remains a gap between theoretical studies and practical usage. Commonly, most SNARKs focus on the field arithmetic, which means that statements are modeled as arithmetic circuits over a finite field. While for real-life applications, typically statements would be originally represented by ring arithmetic. For instance, in verifiable computation, a weak client is more likely to delegate the work of evaluating large computer programs, which are composed of binary operations and $\mathbb{Z}_{2^{64}}$-arithmetic operations.

A direct solution is to emulate binary operations by large field operations, so that SNARKs for fields can be applied. However, this would introduce a significant overhead due to that not only the field size would become much larger, but also the prover needs to additionally show that the computations are actually "operated" over the binary field. Although recent works [18,21,27] have significantly mitigated such overhead by introducing sophisticated embedding techniques into the context, their approaches are restricted to the binary field $\mathbb{F}_2$ and translating $\mathbb{Z}_{2^k}$ operations into binary operations would incur an inherent $O(k^2)$ overhead in the circuit size.

Notably, zero-knowledge virtual machine (zkVM) is a kind of SNARKs developed for computer programs. Roughly speaking, this paradigm first translates computer programs to circuits that "compute" each step of CPU executions, then SNARKs for lookup tables are applied to emulate the CPU instructions. The recent progress of [7,49] has greatly improved the efficiency of zkVMs, in the sense that the costs of emulating one RISC-V instruction are only dominated by committing to six 256-bit field elements. However, as shown in [7], typically the resulting circuits are substantially larger than necessary, constituting a major bottleneck for practical usage. For instance, ECDSA signature verification translates to millions of RISC-V instructions [7].

In contrast, SNARKs designed for $\mathbb{Z}_{2^k}$ arithmetics naturally avoid the expensive translation of the statements. To our best knowledge, the only known con-

[1] Namely, any coin tossed by the verifier is made visible to the prover as soon as it is tossed.

struction is Rinocchio [33], which follows the paradigm of linear probabilistically checkable proofs (linear PCP), and features a constant (in the size of statement) proof size. However, there are typical downsides inherited from the paradigm, i) large prover computation, ii) designated-verifier, iii) trusted setups. Though it has been shown that linear PCPs can be turned into publicly-verifiable SNARKs by using pairings [12,36,44], it remains unclear in the case of rings. Moreover, the concrete efficiency of Rinocchio is unclear as well.

1.1 Our Contributions

Towards the goal of $\mathbb{Z}_{2^k}$, we examine methods of constructing SNARKs over fields, each having non-trivial barriers to overcome. In particular, for the modular routine of basing SNARKs on polynomial IOPs, a major obstacle at first glance is that $\mathbb{Z}_{2^k}$ is not compatible with univariate Lagrange interpolations. Concretely, the largest exceptional set[2] of $\mathbb{Z}_{2^k}$ is of size only 2, meaning that we can only interpolate a linear polynomial over $\mathbb{Z}_{2^k}$. However, polynomial IOPs usually require a sufficiently large exceptional set to interpolate e.g., on the witness. To this end, we resort to Galois rings, the ring extensions of integer rings, which contain larger exceptional sets by setting a larger extension degree.

At the heart of our technical contribution is the proximity result for Reed-Solomon codes over Galois rings, which is crucial for our polynomial commitment schemes (PCS). Then combined with polynomial IOPs for $\mathbb{Z}_{2^k}$, we obtain publicly verifiable SNARKs over $\mathbb{Z}_{2^k}$ and evaluate the concrete efficiency.

Proximity Gaps for Reed-Solomon Codes Over Galois Rings. The proximity problem is first formalized by Ben-Sasson et al. [11] for linear codes over finite fields, and plays a crucial role in analyzing the soundness of code-based PCS. We systematically study the problem for Galois ring cases. Let $\mathsf{GR} = \mathsf{GR}(p^k, d)$ denote a Galois ring extension of $\mathbb{Z}_{p^k}$ with extension degree d. Let C be a general linear error-correction code over GR, where Reed-Solomon (RS) codes constitute a well-studied subclass of such codes. Informally, the proximity problem for C is to determine δ such that: given a batch of words $u_1, \ldots, u_s$, where at least one word that is relatively δ-far from C, then a random linear combination of $u_1, \ldots, u_s$ is still relatively δ-far from C with high probability. We refer δ as the proximity parameter.

In summary, we extend proofs for proximity gaps of $(1-\rho)/4$ and $(1-\rho)/3$ from [3] to Galois rings, where ρ is the code rate of C. In addition, inspired by $(1-\rho)/2$ analysis specially for RS codes in [11], we generalize both the Berlekamp–Welch decoding algorithm and the Polishchuk–Spielman Lemma [45] to work for RS codes over Galois rings. These together allow us to obtain a $(1-\rho)/2$ proximity gap for RS codes over Galois rings. We leave the Johnson radius gap from Guruswami–Sudan list decoding algorithms [37] as future work.

Theorem 1.1 (informal, proximity gap of $(1-\rho)/2$). *Assume $\delta \leq (1-\rho)/2$. Given s vectors $u_1, \ldots, u_s$ over GR, then either all $\{\sum_{i=1}^{s} \gamma^i u_i\}_{\gamma \in \mathsf{GR}/p\mathsf{GR}}$ are*

[2] A set of a commutative ring is exceptional, if the difference of each pair is invertible.

relatively δ-close to RS*, or most of* $\{\sum_{i=1}^{s}\gamma^i u_i\}_{\gamma\in\mathsf{GR}/p\mathsf{GR}}$ *are at least relatively δ-far from* RS*. In the former case,* $u_1,\ldots,u_s$ *are themselves δ-close to* RS*.*

PCS for Galois Rings and Polynomial IOPs for $\mathbb{Z}_{2^k}$. A polynomial commitment for Galois rings allows the prover to first commit to a polynomial $f \in \mathsf{GR}[x]$, and later open an evaluation $f(\alpha)$ on the verifier's choice $\alpha \in \mathsf{GR}$. We adapt two PCS constructions based on Reed-Solomon (RS) codes, from existing paradigms for fields [3,10] to Galois rings. Our first PCS is based on the well-known fast Reed-Solomon IOP of proximity (FRI) [10]. The difficulty here lies in finding suitable $O(1)$-smooth[3] subgroups of Galois rings $\mathsf{GR}(2^k, d)$. Our second PCS is based on Ligero [3]. In general, the efficiency of our PCS for Galois rings is analogue to that from finite fields. Specifically, the FRI-based approach has a smaller proof size, while the Ligero-based approach offers faster proving and flexibility for polynomials of larger degrees.

For polynomial IOPs, we consider statements characterized by a popular language, *rank-one constraint systems* (R1CS) over $\mathbb{Z}_{2^k}$. Specifically, given public $A, B, C \in \mathbb{Z}_{2^k}^{n_1\times n_2}$ and $x \in \mathbb{Z}_{2^k}^{n_0}$, the prover holds a witness $w \in \mathbb{Z}_{2^k}^{n_2-n_0}$, such that $Az * Bz = Cz$, where $z := (x, w) \in \mathbb{Z}_{2^k}^{n_2}$, and $*$ denotes the entry-wise multiplication. Observing that R1CS actually are degree-4 relations on input A, B, C, z, we make use of degree-4 reverse multiplication-friendly embeddings (RMFEs) [20,25,28], turning them into smaller, equivalent, related relations over Galois rings $\mathsf{GR}(2^k, d)$. Note that the introduction of RMFEs allows good concrete efficiency of the resulting SNARKs over $\mathbb{Z}_{2^k}$, as the statement size would be reduced by an $O(\lambda)$ factor, where λ is the security parameter.

Theorem 1.2 (informal, polynomial IOP for $\mathbb{Z}_{2^k}$). *Assume degree-4 RMFEs, there exists a polynomial IOP for $\mathbb{Z}_{2^k}$ with prover computation $\widetilde{O}(N)$, verifier computation $O(N)$, query complexity $O(1)$, oracle size $O(N)$ and soundness error $O(\frac{n_1+n_2}{2^d})$, where N is the number of non-zero entries of the R1CS statement.*

Implementation of SNARKs for Rings. Combining our polynomial IOP and PCS, we obtain SNARKs for rings $\mathbb{Z}_{2^k}$, which are publicly verifiable and do not require a trusted setup. We then implement our constructions based on the NTL library. In particular, RMFEs over Galois rings are concretely instantiated for the first time, and can find many applications [26,29,30,40,41] in secure multi-party computation (MPC) that involve RMFEs. Our experimental evaluations indicate that the running time of our PCSs and IOPs with respect to Galois rings $\mathsf{GR}(2^{16}, d)$ is approximately 2×-4× slower than $\mathsf{GR}(2, d) = \mathbb{F}_{2^d}$. Frankly, there is much room for improving the concrete efficiency and the barrier mainly comes from a lack of fast implementations for Galois ring operations.

1.2 Technical Overview

Proximity Gaps for RS Codes Over Galois Rings. Let $\mathsf{GR} = \mathsf{GR}(p^k, d)$ be a Galois ring and T be the maximal exceptional set of GR with size $|T| = p^d$

[3] We say a set is s-smooth, if its cardinality is some power to s.

(see Sect. 2 for details). Let $\mathsf{RS} = \mathsf{RS}[E, \rho]$ be the Reed-Solomon code defined over an exceptional subset $E \subsetneq T$ with size $|E| = \ell$ and rate ρ. For any vector $w \in \mathsf{GR}^\ell$, let $\mathfrak{d}(w, \mathsf{RS})$ denote the relative distance of w to RS. Firstly, let us give the formal definition of proximity gaps for RS codes over GR, which is similar to [11, Definition 1.1].

Definition 1.1. *Assume $u_0, u_1, \ldots, u_s$ are $s+1$ vectors in GR^ℓ. Let $w(z) := \sum_{i=0}^{s} u_i z^i$. Then the collection curve$(\mathbf{w}) := \{w(\gamma) \in \mathsf{GR}^\ell \mid \gamma \in T\}$ displays a (δ, ϵ)-proximity gap with respect to $\mathsf{RS}[E, \rho]$ if one of the following holds*

1. $\Pr_{\gamma \in T}[\mathfrak{d}(w(\gamma), \mathsf{RS}) \leq \delta] = 1$.
2. $\Pr_{\gamma \in T}[\mathfrak{d}(w(\gamma), \mathsf{RS}) \leq \delta] \leq \epsilon$.

In other words, if there exists $S \subset T$ with size $|S| > \epsilon p^d$ such that for any $\gamma \in S$, the word $w(\gamma) := \sum_{i=0}^{s} u_i \gamma^i$ is relatively δ-close to RS. Then $S = T$. The proofs for $\delta < (1-\rho)/3$ gap with $s = 1$ can be naturally adapted from fields [3] to Galois rings. In fact, this adaptation holds for any general linear code. The major difference is that linear codes defined over Galois rings are *modules* over GR, rather than linear space in the field cases. Though modules and linear spaces share similarities, analyzing modules is generally more complicated.

Generalizing the state-of-the-art $\delta = (1-\rho)/2$ gap shown in [11] to Galois rings is much more challenging. We first sketch the high-level idea of [11] as follows:

Recap the Proof of $(1-\rho)/2$ Gap for Finite Fields. Actually, in [11], aside from the proximity property, all $u_0, u_1, \ldots, u_s$ are proved to be δ-close to the RS code. We also prove an analogue version of $(1-\rho)/2$-correlated agreement over GR.

The main techniques used in the analysis are the Berlekamp–Welch (BW for short) algorithm and Polishchuk–Spielman (PS for short) Lemma. Given a word $w \in \mathbb{F}_q^\ell$ with the relative distance $\mathfrak{d}(w, \mathsf{RS}) \leq \delta$, the Berlekamp–Welch algorithm allows to find the unique codeword $f \in \mathsf{RS}$ satisfying $\mathfrak{d}(w, f) \leq \delta$ by solving the following linear equations:

$$g(\alpha) \cdot w(\alpha) = h(\alpha), \text{ for all } \alpha \in E, \tag{1}$$

where $g(x) = \sum_{i=0}^{e} g_i x^i$, $h(x) = \sum_{i=0}^{\rho\ell+e-1} h_i x^i$ with $\{g_0, \ldots, g_e, h_0, \ldots, h_{\rho\ell+e-1}\}$ to be determined, and $e = \lfloor \delta\ell \rfloor$. The above Eq. (1) are always solvable and let $g(x)$ and $h(x)$ be a tuple of solutions. Then the corrected codeword is $f(x) = h(x)/g(x)$.

Ben-Sasson et al. [11] propose a novel generalization of the BW algorithm from $\mathbb{F}_q$ to the rational function field $\mathbb{F}_q(z)$. Consider the BW decoder with input $w(z) \in \mathbb{F}_q(z)^\ell$, and unknowns $\{A_0(z), \ldots, A_e(z), B_0(z), \ldots, B_{e+\rho\ell-1}(z)\}$, i.e.,

$$A(\alpha, z) w(\alpha, z) = B(\alpha, z), \text{ for all } \alpha \in E, \tag{2}$$

where $A(x, z) = \sum_{i=0}^{e} A_i(z) x^i$ and $B(x, z) = \sum_{i=0}^{\rho\ell+e-1} B_i(z) x^i$. By the condition $|S| \geq \epsilon |\mathbb{F}_q|$, the above equations are also solvable and have a tuple of solutions $A, B \in \mathbb{F}_q[x, z]$ satisfying

$$\deg_z(A) \leq se, \ \deg_z(B) \leq s(e+1).$$

Note that for all $\gamma \in S$, $(A(x,\gamma), B(x,\gamma))$ is also a tuple of solutions of the BW algorithm with input $w(\gamma)$. Namely, there exists a polynomial $P_\gamma(x) \in \mathbb{F}_q[x]_{<\rho\ell}$ such that

$$A(x,\gamma)P_\gamma(x) = B(x,\gamma). \tag{3}$$

Finally, by Eqs. (2) and (3), the Polishchuk–Spielman Lemma implies that $A(x,z) \mid B(x,z)$ if one more condition holds, i.e.,

$$\frac{\deg_x(B)}{\ell} + \frac{\deg_z(B)}{|S|} < 1. \tag{4}$$

Moreover, let $P(x,z) = B/A$. Then

$$\deg_x(P) \le \deg_x(B) - \deg_x(A) \le \rho\ell - 1, \text{ and } \deg_z(P) \le \deg_z(B) - \deg_z(A) \le s.$$

Assume $P = \sum_{i=0}^{s} v_i(x) z^i$. Hence, by Eq. (2), for any $\alpha \in E$ with $A(\alpha, z) \neq 0$, there is

$$P(\alpha, z) = w(\alpha, z) \Rightarrow u_i(\alpha) = v_i(\alpha), \quad \text{for all} \quad i = 0, 1, \ldots, s.$$

Consequently, for all $\gamma \in \mathbb{F}_q$, $\mathfrak{d}\left(P(x,\gamma), w(x,\gamma)\right) \le \delta$, and for all $i = 0, 1, \ldots, s$, $\mathfrak{d}\left(v_i, u_i\right) \le \delta$.

Overview Our Proof of $(1-\rho)/2$ *Gap Over Galois Rings.* To generalize the above proof to RS codes over Galois rings GR, there stand the following three main technique issues:

(i) Keep the main properties of Reed-Solomon codes.
(ii) Show that the above Eq. (2) are still solvable over the polynomial ring $\mathsf{GR}[z]$.
(iii) Formalize the PS Lemma for GR and prove its correctness.

As we will see in Sect. 4.1, by setting the evaluation points set E as an exceptional subset of GR, many useful properties of RS codes over fields are preserved, hence sufficient for applying our analysis. Similarly, given $u_0, u_1, \ldots, u_s$, we view the function $w(z) = \sum_{i=0}^{s} u_i z^i$ as an RS codeword over $\mathsf{GR}[z]$. In the field case, by using Cramer's rule over the rational function field $\mathbb{F}_q(z)$, the BW Eq. (2) are solvable and produce two bivariate polynomials $A(x,z)$ and $B(x,z)$ with bounded degrees. However, in the Galois ring case, since $\mathsf{GR}[z]$ is not a domain, Cramer's rule is no longer applicable. Instead, we first use a theory proposed by McCoy [42] to show the solvability of BW equations over $\mathsf{GR}[z]$. Then by further exploiting the properties of the Galois rings, we eventually find the desired solution (A, B) with bounded degrees.

Now, assume $A, B \in \mathsf{GR}[x,z]$ is a tuple of solutions. Then, Eq. (2) and Eq. (3) still hold for A, B with all $\alpha \in E$ and $\gamma \in S$. Note that the PS lemma [11] is only applicable to the field case. To adapt it to the Galois ring case, we have first to formalize it over GR and prove its correctness. The novelty of our adaptation is that we use the p-adic lifting technique in our proof, which clearly is not necessary for the original proof in the field case. Consequently, we rewrite the condition (4) in PS Lemma for GR as follows

$$\frac{\deg_x(B)}{\ell - e} + \frac{\deg_z(B)}{|S| - se} < 1, \tag{5}$$

where $e = \lfloor \delta\ell \rfloor$. Finally, we get a polynomial $P(x,z) = \sum_{i=0}^{e} v_i(x)z^i \in \mathsf{GR}[x,z]$ with $\deg_x(v_i(x)) \leq \rho\ell - 1$ for all $i = 0, 1, \ldots, s$, which satisfies that, for all γ in the maximal exceptional set $T \subset \mathsf{GR}$, $\mathfrak{d}\left(P(x,\gamma), w(x,\gamma)\right) \leq \delta$, and $\mathfrak{d}\left(v_i, u_i\right) \leq \delta$ for all $i = 0, 1, \ldots, s$.

It is worth mentioning that, to ensure the condition (5) holds for $e \leq (1-\rho)/2$, we need $|S| = O(\ell^2)$ which implies that the code length ℓ is less than $\sqrt{|T|} = 2^{d/2}$. For a linear size $\ell = O(2^d)$, then $|S|$ is at most $O(\ell)$, leading to a $(1-\rho)/3$ gap for RS codes by the modified condition (5) over GR. Our analyses can be naturally generalized to any $\mathsf{GR}(p^k, d)$.

IOPs for $\mathbb{Z}_{2^k}$ via Degree-4 RMFEs. A degree-D RMFE over $\mathbb{Z}_{2^k}$ consists of two $\mathbb{Z}_{2^k}$-linear maps, $\phi : \mathbb{Z}_{2^k}^m \to \mathsf{GR}(2^k, d)$, and $\psi : \mathsf{GR}(2^k, d) \to \mathbb{Z}_{2^k}^m$, such that

$$\psi\big(\phi(\mathbf{x}_1) \cdot \cdots \cdot \phi(\mathbf{x}_D)\big) = \mathbf{x}_1 * \cdots * \mathbf{x}_D,$$

for any $\mathbf{x}_1, \ldots, \mathbf{x}_D \in \mathbb{Z}_{2^k}^m$, where m is some positive integer and $*$ denotes the entry-wise multiplication. Informally, RMFEs allow to convert computations over $\mathbb{Z}_{2^k}$ of a limited multiplication depth into some equivalent computations over $\mathsf{GR}(2^k, d)$. Observe that an R1CS relation $A, B, C \in \mathbb{Z}_{2^k}^{n_1 \times n_2}, z \in \mathbb{Z}_{2^k}^{n_2}$ with $Az * Bz = Cz$ is essentially a degree-4 relation, we can employ degree-4 RMFEs for the transformation. Let $\hat{A}$ denote the $n_1 \times \frac{n_2}{m}$ matrix over GR obtained by applying ϕ on every block (of size m) of each row of A accordingly. We define $\hat{B}, \hat{C}$ and $\hat{z} \in \mathsf{GR}^{\frac{n_2}{m}}$ in a similar way. Then the R1CS relation over $\mathbb{Z}_{2^k}$ is equivalent to the following:

$$\psi(\hat{A}\hat{z} * \hat{B}\hat{z}) = \psi(\hat{C}\hat{z}). \tag{6}$$

To efficiently check Eq. (6) via GR arithmetic, we further rewrite it by using algebraic properties of RMFEs. According to [28], we can assume $\phi(\mathbf{1}) = 1$, where $\mathbf{1}$ denotes all one's vector, and basically we have that the direct sum of the image of ϕ and the kernel of ψ is exactly the GR. Therefore, now we can rewrite the relation by introducing an $\eta \in \mathrm{Ker}(\psi)^{n_1}$, such that

$$\hat{A}\hat{z} * \hat{B}\hat{z} = \hat{C}\hat{z} + \eta. \tag{7}$$

Since z is held and encoded by the prover, $\hat{z}$ should be guaranteed to be in $\mathrm{Im}(\phi)$. Through the characterization of Eq. (7), we can reduce R1CS relations over $\mathbb{Z}_{2^k}$ to Rowcheck, Lincheck relations for GR as usual IOPs do, and in addition specialized Imcheck, Kercheck for probabilistically checking whether a vector is over $\mathrm{Im}(\phi)$ or over $\mathrm{Ker}(\psi)$. By extending the approaches of [13,21] from fields to Galois rings, we construct polynomial IOPs for the above four kinds of relations, yielding polynomial IOPs for R1CS over $\mathbb{Z}_{2^k}$. We remark that one can distill a reduction via degree-2 RMFEs from [21], which also leads to an IOP for R1CS over $\mathbb{Z}_{2^k}$. While, our reduction by degree-4 RMFEs is conceptually simpler and admits SNARKs of smaller argument size.

1.3 Related Work

Concurrently, two independent works [38,54] construct PCS over Galois rings using Spielman's linear time encodable codes [50], and proves the $\frac{1-\rho}{4}$, $\frac{1-\rho}{3}$

proximity gaps for general linear codes over Galois rings. Additionally, recent works [34,55] prove the 1.5 Johnson radius gap for linear codes, and it is principally possible to extend their analyses to Galois rings. We instead build upon Reed-Solomon codes and prove the $\frac{1-\rho}{2}$ proximity gap for RS codes over Galois rings, which admits smaller proof sizes.

Recently, a line of works [1,2,22,24,31,39,43,53] construct PCS from lattice, among which Greyhound [43] has the best performance, and is claimed to have much better concrete efficiency than Ligero or FRI-based approaches. It remains very interesting whether these lattice-based approaches can be generalized to Galois rings. Concurrently, Arnon et al. propose optimizations of FRI [4–6], which significantly improve the efficiency of FRI. In particular, extending STIR [4] to Galois rings seems possible, and we leave it as a future work. Also, there is a line of works [8,9,17,40] focusing on realizing concretely efficient zero-knowledge (ZK) protocols for rings $\mathbb{Z}_{2^k}$. In general, these works follow the paradigm of "commit-and-prove", and exploit techniques from secure multiparty computations (MPC). However, these works have linear proof sizes, hence not *succinct*.

2 Preliminaries

Interactive Oracle Proofs. Interactive oracle proofs [14,47] generalize both interactive proofs (IP) and probabilistically checkable proofs (PCP), in the sense that the verifier can accept or reject the proof without entirely reading every prover's message in each round. Namely, the verifier has query access to the so-called "oracle" messages, and he can query any symbol at any time. Ben-Sasson et al. [14] propose a compiler that allows transforming any IOP into a non-interactive argument in the random oracle model, via Merkle-tree hashing and Fiat-Shamir transformation. This allows for obtaining SNARKs by designing efficient IOPs with few queries.

Polynomial IOP is a variant of IOP, in which each oracle message specifies a polynomial, and the verifier can query the polynomial at any point, then get the evaluation. IOP of proximity (IOPP) allows the verifier to test whether an oracle message (specifying a vector) is close to any codeword of a given code.

Polynomial Commitments. A common way to build SNARKs is to combine polynomial IOPs and polynomial commitments. Roughly speaking, a polynomial commitment allows the prover to commit to (all evaluations of) a polynomial f of a bounded degree, and later reveal $f(\alpha)$ on the verifier's choice. The commitment size should be at least sublinear in the number of all possible evaluations, so that the polynomial commitment scheme is non-trivial. The scheme satisfies *hiding*, where the verifier learns nothing before the evaluation phase; and *binding*, where the prover is effectively forced to reveal $f(\alpha)$ for a fixed polynomial f determined at the time of the original commitment. Polynomial commitments can be constructed from IOPPs, where the oracles are then instantiated with Merkle-tree hashes via the BCS compiler [14].

Galois Rings. This section introduces key definitions and properties of Galois rings that will be essential for our study. We refer the reader to [52] for more details. Let p be a prime and $\mathbb{Z}_{p^k}$ be the integer residue classes modulo p^k. The Galois ring is an extension of $\mathbb{Z}_{p^k}$ and has characteristic p^k. Specifically, let $d \geq 1$ be an integer and $f(x) \in \mathbb{Z}_{p^k}[x]$ be a monic polynomial of degree d such that $\bar{f}(x) := f(x) \bmod p$ is an irreducible polynomial over the finite field $\mathbb{F}_p$. The Galois ring of extension degree d is defined as

$$\mathsf{GR}(p^k, d) = \mathbb{Z}_{p^k}[x]/(f(x)).$$

Let $\xi = x + (f(x)) \in \mathsf{GR}(p^k, d)$ be a root of $f(x)$. Then any $\alpha \in \mathsf{GR}(p^k, d)$ can be expressed uniquely in the form of $\alpha = a_0 + a_1\xi + \cdots + a_{d-1}\xi^{d-1}$, $a_i \in \mathbb{Z}_{p^k}$.

Proposition 2.1. *[52]*

(1) $\mathsf{GR}(p^k, d)$ *is a local principal ideal ring with the maximal ideal* (p)*. Moreover, there is a field isomorphism* $\tau :\ \mathsf{GR}(p^k, d)/(p) \cong \mathbb{F}_{p^d} = \mathbb{F}_p[x]/(\bar{f}(x))$*.*

(2) There exists an element $\zeta \in \mathsf{GR}(p^k, d)$ *of multiplicative order* $p^d - 1$*. Moreover, let* $T = \{0, 1, \zeta, \zeta^2, \ldots, \zeta^{p^d-2}\}$*. Then* T *is a set of representatives for equivalence classes modulo* (p)*. Hence any* $\alpha \in \mathsf{GR}(p^k, d)$ *can be also expressed uniquely in the form of* $\alpha = \alpha_0 + \alpha_1 p + \cdots + \alpha_{k-1}p^{k-1}$*,* $\alpha_i \in T$ *and* α *is a unit if and only if* $\alpha \bmod p \neq 0$*.*

(3) T *is an exceptional set, i.e., the difference of any two elements in* T *is a unit in* $\mathsf{GR}(p^k, d)$*.*

From now on, we sometimes simply denote $\mathsf{GR}(p^k, d)$ by GR. The following lemma shows that the multi-point evaluation of a polynomial $f(x)$ in $\mathsf{GR}[x]$ at an exceptional set of size greater than $\deg(f)$ uniquely determines itself.

Lemma 2.1 ([46, Corollary 9]). *Let* $f(x)$ *be a nonzero polynomial over* GR *of degree* r *and* $r \leq p^d$*. Then* $f(x)$ *has at most* r *roots in an exceptional set of* GR*.*

From above, we have a Galois ring analogue of the Schwartz-Zippel Lemma for bounding the soundness error.

Lemma 2.2 (Schwartz-Zippel Lemma). *Let* $f(x)$ *be a non-zero polynomial over* $\mathsf{GR}(p^k, d)$ *of degree* $\leq r$ *and* T *be the exceptional set as in Proposition 2.1. Then*

$$\Pr\left[f(\alpha) = 0 : \alpha \xleftarrow{\$} T\right] \leq \frac{r}{p^d}.$$

The next Lemma allows to do Lagrange interpolation when the points at which interpolation is done are well chosen.

Corollary 2.1 ([46, Corollary 10]). *Let* $E = \{\alpha_1, \ldots, \alpha_n\}$ *be an exceptional subset of* GR *and* $\{\beta_1, \ldots, \beta_n\}$ *be a subset of* GR*. Then there exists a unique polynomial* $f \in \mathsf{GR}[x]$ *of degree at most* $n - 1$ *such that* $f(\alpha_i) = \beta_i$*, for* $i = 1, \ldots, n$*.*

Error Correcting Codes over Galois Ring. As in the linear case of error-correcting codes over finite fields, we define linear error-correcting codes of code C length n as submodules of GR^n. Let us first recap the definition of modules over a ring.

Definition 2.1. *A module M over a ring $\mathcal{R}$ is a set of objects that is an abelian group under addition and is closed under multiplication by scalars from $\mathcal{R}$. The operation of scalar multiplication is distributive and associative.*

Definition 2.2. *Suppose M is an R-module and $\{M_i : i \in I\}$ is a series of submodules of M. If every x in M can be written in exactly one way as a sum of finitely many elements of the M_i, then we say that M is the internal direct sum of the submodules M_i, denoted by $M = \bigoplus_{i\in I} M_i$.*

A submodule N of M is a direct sum of M if there exists some other submodule N' of M such that M is the internal direct sum of N and N'.

For error-correction codes over GR, we have the following properties, which will be used in proving the $(1-\rho)/3$ gap for general linear codes over GR. The proof is deferred to the full version.

Lemma 2.3. *Assume $M \subset \mathsf{GR}(p^k, d)^n$ is a submodule. For any nonzero element $u \in M$, then $\mathsf{GR}(p^k, d)u$ is a direct sum of M, i.e., every $x \in M$ can be uniquely expressed as $x = \alpha u + v$ for some $\alpha \in \mathsf{GR}$ and $v \in M$.*

For any $u \in \mathsf{GR}^n$, we call the Hamming weight of u the number of nonzero entries of u and denote it by $\mathrm{wt}(u)$. Let $C \subset \mathsf{GR}^n$ be a linear error-correcting code. The minimum Hamming distance of C denoted by D_C is defined as $D_C = \min_{0\neq c\in C}\{\mathrm{wt}(c)\}$. The relative minimum distance is defined as $\mathfrak{d}_C := D_C/n$.

Similarly, the Hamming distance and relative Hamming distance of a vector $u \in \mathsf{GR}^n$ and C are respectively defined as

$$\mathsf{d}(u, C) = \min_{c\in C}\{\mathrm{wt}(u-c)\}, \text{ and } \mathfrak{d}(u, C) = \mathsf{d}(u, C)/n.$$

Furthermore, assuming $v \in C$ is the codeword closest to u, we denote the $\Delta(u, C)$ as the support set of $u - v$, i.e.,

$$\Delta(u, C) := \mathrm{supp}(u - v) := \{i \in [n] : u_i \neq v_i\}.$$

Definition 2.3 (Reed-Solomon codes). *Let $E = \{\alpha_1, \ldots, \alpha_n\} \subset \mathsf{GR}(p^k, d)$ be an exceptional set of size n and $\rho \in (0, 1]$ be a real number. Let $\mathsf{GR}[x]_{<\rho n}$ be the set of polynomials of degree less than ρn. The Reed-Solomon code over $\mathsf{GR}(p^k, d)$ is defined as follows*

$$\mathsf{RS}[E, \rho] = \{\mathrm{ev}_E(f) := (f(\alpha_1), \ldots, f(\alpha_n)) \mid f \in \mathsf{GR}[x]_{<\rho n}\}.$$

Assume m real numbers $\rho_1 \ldots \rho_m$ in $(0, 1]$, the interleaved RS codes is defined as

$$\mathsf{RS}[E,(\rho_1,...,\rho_m)] := \left\{ \begin{pmatrix} \mathrm{ev}_E(f_1) \\ \mathrm{ev}_E(f_2) \\ \vdots \\ \mathrm{ev}_E(f_m) \end{pmatrix} \mid f_i(x) \in \mathsf{GR}(p^k,d)[x]_{<n\rho_i},\ i=1,\ldots,m \right\}.$$

If $\rho_1,...,\rho_m$ are identical, we simply write it as $\mathsf{RS}^m[E,\rho]$.

From Lemma 2.1, $\mathsf{RS}[E,\rho]$ has minimum distance $D = n-\rho n+1$. We remark that $\mathsf{RS}[E,\rho]$ can also be viewed as a set of functions $\{f : E \to \mathsf{GR}\}$.

For a matrix $\mathsf{G} \in \mathsf{GR}^{m\times n}$, let $\mathsf{G}[j]$ denote the j-th column of G. Similarly, the minimum distance between G and the interleaved RS code RS^m is defined as

$$d(\mathsf{G},\mathsf{RS}^m) = \min_{\mathsf{C}\in\mathsf{RS}^m}\{d(\mathsf{G},\mathsf{C})\} := \min_{\mathsf{C}\in\mathsf{RS}^m}|\{j\in[n] \mid M[j]\neq \mathsf{C}[j]\}|.$$

Degree-D RMFE. Degree-D Reverse Multiplication Friendly Embeddings [20,25,28] (RMFE) are powerful mathematical tools that allow to transform computations over a small Galois ring $\mathsf{GR}(p^k,r)$ with a bounded multiplication depth to computations over a large Galois ring extension $\mathsf{GR}(p^k,rd)$.

Definition 2.4 Degree-D RMFE). *Let p be a prime, $k,r,m,d,D \geq 1$ be integers. A pair (ϕ,ψ) is called an $(m,d;D)$-RMFE over $\mathsf{GR}(p^k,r)$, if $\phi : \mathsf{GR}(p^k,r)^m \to \mathsf{GR}(p^k,rd)$ and $\psi : \mathsf{GR}(p^k,rd) \to \mathsf{GR}(p^k,r)^m$ are two $\mathsf{GR}(p^k,r)$-linear maps such that*

$$\psi(\phi(\boldsymbol{x}_1)\cdot\phi(\boldsymbol{x}_2)\cdots\phi(\boldsymbol{x}_D)) = \boldsymbol{x}_1 * \boldsymbol{x}_2 * \cdots * \boldsymbol{x}_D,$$

for all $\boldsymbol{x}_1,\boldsymbol{x}_2,...,\boldsymbol{x}_D \in \mathsf{GR}(p^k,r)^m$, where $*$ denotes the entry-wise multiplication.

In this paper we always assume $\phi(\mathbf{1}) = 1$, where $\mathbf{1}$ denotes the vector of all 1's, since as shown in [28], such $(m,d;D)$-RMFEs always exist. Then RMFEs have the following properties.

Lemma 2.4. *[41, Lemma 2]. Let (ϕ,ψ) be an $(m,d;D)$-RMFE over Galois ring $\mathsf{GR}(p^k,r)$, then $\mathsf{GR}(p^k,rd)$ is the direct sum of $\mathrm{Ker}(\psi),\mathrm{Im}(\phi)$. That is $\mathsf{GR}(p^k,rd) = \mathrm{Ker}(\psi)\oplus\mathrm{Im}(\phi)$.*

Lemma 2.5. *[28, Corollary 5]. There exists a family of $(m,d;D)$-RMFEs over $\mathbb{Z}_{2^k}$ for all $k\geq 1$ with $m\to\infty$ and $\frac{d}{m} \to \frac{1+2D}{3}(D+\frac{D(3+1/(2^D-1))}{2^{D+1}-1})$.*

For instance, when $m\to\infty$, there exists a family of $(m,d;2)$-RMFEs over $\mathbb{Z}_{2^k}$ with $\frac{d}{m}\to 4.92$, and a family of $(m,d;4)$-RMFEs over $\mathbb{Z}_{2^k}$ with $\frac{d}{m}\to 13.18$.

3 Proximity Gaps for Reed-Solomon Codes Over Galois Rings

Informally, the proximity problem is to determine δ such that: given a batch of words $u_1, \ldots, u_s$, where at least one word that is relatively δ-far from RS, then a random linear combination of $u_1, \ldots, u_s$ is still relatively δ-far from RS with high probability. We refer δ as the proximity parameter. We systematically study the proximity problem for RS codes defined over Galois rings, which is defined formally in Definition 1.1. A notable difference compared to the field case is that the span of $c_1, \ldots, c_n$ is a *module* over GR rather than a vector space over $\mathbb{F}$.

For completeness, we first establish two proximity gap results for $\delta \leq (1-\rho)/4$ and $\delta \leq (1-\rho)/3$, where ρ is the rate of RS over Galois rings, following similar analyses from [3]. We then employ the state-of-the-art analysis from [11], and prove a $(1-\rho)/2$ gap for Reed-Solomon codes over Galois rings. Specifically, our results are obtained by generalizing the Berlekamp–Welch decoding algorithm and the Polishchuk–Spielman Lemma to Galois rings. Let $\mathsf{RS} = \mathsf{RS}[E, \rho]$ denote a Reed-Solomon code defined over an exceptional $E \subset \mathsf{GR} = \mathsf{GR}(p^k, d)$ with code rate ρ. Let D be the minimal distance of RS, and $\ell = |E|$.

3.1 $(1-\rho)/4$ and $(1-\rho)/3$ Gaps for Linear Codes

The following theorem implies the $(1-\rho)/4$ gap results for RS codes over GR. We refer to the full version of this paper for a formal proof. The analysis actually applies to any linear code over Galois rings $\mathsf{GR}(p^k, d)$.

Theorem 3.1. *Let e be a positive integer such that $e < D/4$, and $G^* \in \mathsf{GR}^{m \times \ell}$. Suppose $\mathsf{d}(G^*, \mathsf{RS}^m) > e$. Then, for a random u^* in the row-span of G^*, we have*

$$\Pr\left[\mathsf{d}(u^*, \mathsf{RS}) \leq e\right] \leq \frac{e+1}{p^d}.$$

We next prove the following theorem, which essentially improves the $(1-\rho)/4$ gap to $(1-\rho)/3$, similarly as the field case. The analysis also applies to any linear code over Galois rings $\mathsf{GR}(p^k, d)$.

Theorem 3.2. *Let e be a positive integer such that $e < D/3$, and $G^* \in \mathsf{GR}^{m \times \ell}$. Suppose $\mathsf{d}(G^*, \mathsf{RS}^m) > e$. Then, for a random u^* in the row-span of G^*, we have*

$$\Pr\left[\mathsf{d}(u^*, \mathsf{RS}) \leq e\right] \leq \frac{D}{p^d}. \tag{8}$$

First, we have the following Lemma, which can be proved similarly as in [3, Lemma 4.7] by combining our Lemma 2.3.

Lemma 3.1. *Let D be the minimal distance of RS over $\mathsf{GR}(p^k, d)$. Let e be a positive integer such that $e < D/3$ and $p^d \geq 2e$. Assume $\mathsf{d}(G^*, \mathsf{RS}^m) > e$. Then, there exists $v^* \in L^*$ such that $\mathsf{d}(v^*, \mathsf{RS}) > e$, where L^* is the row-span of G^*.*

Proof. We prove this by contradiction. Assume $\mathsf{d}(v^*, \mathsf{RS}) \leq e$ for all $v^* \in L^*$ and suppose that $v_0^* \in L^*$ maximizes the distance from RS. Since $\mathsf{d}(G^*, \mathsf{RS}^m) > e$, there must be a row $G_i^* \in L^*$ such that $\Delta(G_i^*, \mathsf{RS}) \backslash \Delta(v_0^*, \mathsf{RS}) \neq \emptyset$, as v_0^* introduces at most e errors by assumption. Let $v_0^* = u_0 + \chi_0$ and $G_i^* = u_i + \chi_i$ for $u_0, u_i \in \mathsf{RS}$, then χ_0, χ_i have weight $\leq e$.

We claim that there exists $\alpha \in \mathsf{GR}$ such that for $\hat{v} = v_0^* + \alpha G_i^*$, we have $\mathsf{d}(\hat{v}, \mathsf{RS}) > \mathsf{d}(v_0^*, \mathsf{RS})$, contradicting to the choice of v_0^*.

First we show that $w = u_0 + \alpha u_i \in \mathsf{RS}$ is the closest to $\hat{v} = v_0^* + \alpha G_i^*$. Since $\mathsf{d}(v_0^*, \mathsf{RS}) \leq e$ and $\mathsf{d}(G_i^*, \mathsf{RS}) \leq e$, $w \in \mathsf{RS}$ is at most $2e$-far from $\hat{v}$. Furthermore, since $\mathsf{d}(\hat{v}, \mathsf{RS}) \leq e$ there must be a codeword $w' \in \mathsf{RS}$ that is at most e-far from $\hat{v}$. Now, since $D > 3e$, it must be the case $w = w'$, otherwise $\mathsf{d}(w, w') \leq 3e < D$.

Therefore, for any $j \in \Delta(G_i^*, \mathsf{RS}) \cup \Delta(v_0^*, \mathsf{RS}) = \mathrm{supp}(\chi_0) \cup \mathrm{supp}(\chi_1)$ there is at most $p^{(k-1)d}$ choices of α such that the j-th component of $\chi_0 + \alpha\chi_i$ vanishes. By a union bound, there are at most $2e \cdot p^{(k-1)d}$ such α's. Considering any other $\alpha \in \mathsf{GR}$ and $\hat{v} = v_0^* + \alpha G_i^*$ (the existence of such α assuring by $p^d \geq 2e$), we have $\Delta(\hat{v}, \mathsf{RS}) \supsetneq \Delta(v_0^*, \mathsf{RS})$, which is a contradiction to the choice of v_0^*. This completes the proof. □

Proof (the proof of Theorem 3.2). Let L^* be the module generated by row vectors of G^*.

CASE 1*:* Assume there exist one $v^* \in L^*$ such that $\mathsf{d}(v^*, L) > 2e$. In this case, by the same analysis as in Lemma 3.1, we can prove that

$$\Pr\left[\mathsf{d}(u^*, \mathsf{RS}) \leq e\right] \leq \frac{1}{p^d}.$$

CASE 2*:* Assume for all $v^* \in L^*$, there are $\mathsf{d}(v^*, L) \leq 2e$. By Lemma 3.1, there exists one $v^* \in L^*$ such that $\mathsf{d}(v^*, L) > e$. Assume $v^* = p^t v_0$ and $v_0 \not\equiv 0 \bmod p$. For a random $u^* \in L^*$, we know that $u^* = \alpha v^* + x$ by Lemma 2.3, where $\alpha \in \mathsf{GR}/p^{k-t}\mathsf{GR}$ and x is distributed independently with α. Conditioned on every possible choice of x, we have that there is at least one α such that $\mathsf{d}(\alpha v^* + x, L) > e$. In this case, we claim that there at most $p^{(k-t-1)d}$ choices of $\alpha \in \mathsf{GR}/p^{k-t}\mathsf{GR}$ such that $\mathsf{d}(\alpha v^* + x, L) \leq e$.

Before the proof of our claim, we note that the number of α such that $\mathsf{d}(\alpha v^* + x, L) \leq e$ is equal to the number of α such that $\mathsf{d}(\alpha(v^* + c) + (x + c'), L) \leq e$ for any $c, c' \in L$. Thus we assume v^* and x both have weight at most $2e$ by our assumption in Case 2[4]. Since $|\mathrm{supp}(v^*) \cup \mathrm{supp}(x)| > e$, the element α such that $\mathsf{d}(\alpha v^* + x, L) \leq e$ must satisfy $|\mathrm{supp}(\alpha v^* + x)| \leq e$. For each coordinate $j \in \mathrm{supp}(v^*) \cap \mathrm{supp}(x)$, there are at most $p^{(k-t-1)d}$ choices of α vanishing the j-th coordinate of $\alpha v^* + x$. As $|\mathrm{supp}(v^*) \cap \mathrm{supp}(x)| \leq 2e$, there are at most $2e \cdot p^{(k-t-1)d}$ choices of α such that $\mathsf{d}(\alpha v^* + x, L) \leq e$. Hence

$$\Pr\left[\mathsf{d}(u^*, \mathsf{RS}) \leq e\right] \leq \frac{2e \cdot p^{(k-t-1)d}}{p^{(k-t)d}} < \frac{D}{p^d}.$$

[4] There is a small issue of [3, Claim A.2], where they claim it suffices to consider each v^*, x has weight at most e without the assumption of CASE 2, which does not make sense. This issue can be fixed by our arguments.

3.2 Beating the $(1-\rho)/3$ Gap for Reed-Solomon Codes

Theorem 3.3. *Assume $\delta \leq (1-\rho)/2$. Given s vectors $u_0, u_1, \ldots, u_s \in \mathsf{GR}(p^k, d)^\ell$, let $S = \{\gamma \in T \mid \mathfrak{d}\left(\sum_{i=0}^{s} u_i \gamma^i, \mathsf{RS}[E, \rho]\right) \leq \delta\}$, where T is the exceptional set of $\mathsf{GR}(p^k, d)$ of order p^d. Suppose $|S| > s\ell^2$. Then $S = T$. Furthermore there are $v_0, v_1, \ldots, v_s \in \mathsf{RS}[E, \rho]$ and a set $E' \subset E$ with size $|E'| \geq \ell(1-\delta)$ such that for all $i = 0, 1, \ldots, s$, v_i agrees with u_i on all points in E'.*

Assume $u_0(x), u_1(x), \ldots, u_s(x)$ be s polynomials of degree less than ℓ which are interpolated from $u_i : E \to \mathsf{GR}(p^k, d)^\ell$ for all i. Let $w(x, z) := \sum_{i=0}^{s} u_i(x) z^i \in \mathsf{GR}[x, z]$. Then w is a function from E to the univariate polynomial ring $\mathsf{GR}[z]$:

$$w : \ \alpha \in E \mapsto \sum_{i=0}^{s} u_i(\alpha) z^i.$$

The proof of Theorem 3.3 depends on the following two Lemmas that are formally proved after the proof of Theorem 3.3.

Lemma 3.2. *Let δ, $u_0, u_1, \ldots, u_s$ and S be defined as in Theorem 3.3. Let $e = \lfloor \delta\ell \rfloor$ and $w(x, z) := \sum_{i=0}^{s} u_i(x) z^i$ a bivariate polynomial over GR. If $|S| > s\ell^2$, then there exist two bivariate polynomials $A(x, z), B(x, z) \in \mathsf{GR}[x, z]$ in the form of $A(x, z) = \sum_{i=0}^{e} A_i(z) x^i, B(x, z) = \sum_{i=0}^{\rho\ell+e-1} B_i(z) x^i$ satisfying the following linear equations over $\mathsf{GR}[z]$:*

$$A(\alpha, z) w(\alpha, z) = B(\alpha, z), \ \textit{for all } \alpha \in E, \tag{9}$$

Furthermore, the degrees of $A(x, z)$ and $B(x, z)$ satisfy

$$\deg_z(A) \leq se, \ \deg_z(B) \leq s(e+1).$$

Lemma 3.3 (Polishchuk–Spielman Lemma over Galois ring). *Let $\mathsf{GR} = \mathsf{GR}(p^k, d)$ be a Galois ring and T be the maximal exceptional set. Assume $A(x, z), B(x, z) \in \mathsf{GR}[x, z]$ be polynomials with degrees bounded by*

$$\deg_x(A) \leq a_x, \ \deg_x(B) \leq b_x, \ \deg_z(A) \leq a_z, \ \deg_z(B) \leq b_z.$$

Suppose that for at least n_x values of $\alpha \in T$, there exist polynomials $M_\alpha(z) \in \mathsf{GR}[z]$ with $\deg_z(M_\alpha(z)) \leq d_z := b_z - a_z$ such that $B(\alpha, z) = M_\alpha(z) A(\alpha, z)$, and similarly suppose that for at least n_z values of $\gamma \in T$, there exist polynomials $N_\gamma(x) \in \mathsf{GR}[x]$ with $\deg_x(N_\gamma(x)) \leq d_x := b_x - a_x$ such that $B(x, \gamma) = N_\gamma(x) A(x, \gamma)$. Finally, suppose

$$\frac{b_x}{n_x - a_x} + \frac{b_z}{n_z - a_z} < 1.$$

If all conditions hold, then $A(x, z) \mid B(x, z)$ as polynomials in $\mathsf{GR}[x, z]$. Furthermore, the quotient $P(x, z) = B/A$ satisfies

$$deg_x(P) \leq d_x, \ \deg_z(P) \leq d_z,$$

as well as $P(x, \gamma) = N_\gamma(x)$ for at least $n_z - a_z$ values of $\gamma \in T$, and $P(\alpha, z) = M_\alpha(z)$ for at least $n_x - a_x$ values of $\alpha \in T$.

Proof (the proof of Theorem 3.3) We will follow the proof of [11, Theorem 4.1], which consists of two main steps:

Step 1. By Lemma 3.2, there exist $A(x,z) = \sum_{i=0}^{e} A_i(z)x^i, B(x,z) = \sum_{i=0}^{e} B_i(z)x^i \in \mathsf{GR}[x,z]$ such that

$$A(\alpha,z)w(\alpha,z) = B(\alpha,z), \text{ for all } \alpha \in E. \tag{10}$$

Furthermore,

$$\deg_x(A) \le e,\ \deg_x(B) \le e+\rho\ell-1,\ \deg_z(A) \le se,\ \deg_z(B) \le s(e+1). \tag{11}$$

By assumptions, for any $\gamma \in S$, since the relative distance $\mathfrak{d}(w(x,\gamma),\mathsf{RS}) \le \delta \le (1-\rho)/2$, by Eqs. (10) and (11), $A(x,\gamma)$ and $B(x,\gamma)$ is a tuple of solutions for the Berlekamp–Welch decoder with input $w(x,\gamma)$. Let $N_\gamma(x) = B(x,\gamma)/A(x,\gamma)$. Then we have the following equations:

$$A(x,\gamma)N_\gamma(x) = B(x,\gamma) \text{ for all } \gamma \in S.$$

Then by PS Lemma 3.3, if

$$\frac{e+\rho\ell-1}{\ell-e} + \frac{s(e+1)}{|S|-se} < 1 \tag{12}$$

then the following Step 2 also holds. We defer the proof of Eq. (12) at the end of this proof.

Step 2. $A(x,z) \mid B(x,z)$ over GR. Let $P(x,z) = B(x,z)/A(x,z)$. Then $\deg_x(P) < \rho\ell$ and $\deg_z(P) \le s$. Assume $P = \sum_{i=0}^{s} v_i(x)z^i$. According to Eq. (10), for $\alpha \in E$ such that $A(\alpha,z) \ne 0$, we have

$$\sum_{i=0}^{s} v_i(\alpha)z^i = P(\alpha,z) = \sum_{i=0}^{s} u_i(\alpha)z^i$$

As $\deg_x(A) \le e = \lfloor \delta\ell \rfloor$, the set $Z := \{\alpha \in E : \ A(\alpha,z) = 0\}$ has size at most $\lfloor \delta\ell \rfloor$. Let $E' = E \setminus Z$ and $|E'| \ge \lceil \ell(1-\delta) \rceil$. Then for any $\alpha \in E'$, $u_i(\alpha) = v_i(\alpha)$ for all i. Namely, the relative distance

$$\mathfrak{d}(u_i, v_i) \le \delta \ \forall\ i = 0, 1, \ldots, s.$$

Furthermore, for $\alpha \in E'$, we have

$$w(\alpha,\gamma) = P(\alpha,\gamma),\ \forall\ \gamma \in T \Rightarrow S = T.$$

To complete the proof, the last thing we need to show is that under the settings $e = \lfloor \delta\ell \rfloor \le (1-\rho)\ell/2$ and $|S| > s\ell^2$, the condition (12) always hold. Note that the function $f(e,|S|) := \frac{e+\rho\ell-1}{\ell-e} + \frac{s(e+1)}{|S|-se}$ is strictly increasing in e and strictly decreasing in $|S|$ within the domain $e < (1-\rho)\ell/2$ and $|S| > s\ell^2$. By setting $e = (1-\rho)\ell/2$ and we solve $|S|$ from Eq. (12), i.e.,

$$\frac{e+\ell\rho-1}{\ell-e} + \frac{s(e+1)}{|S|-se} < 1.$$

Then we get $|S| > s\ell^2(\frac{1-\rho^2}{4} + \frac{1}{\ell})$. Thus, the above inequality always holds for $|S| \ge s\ell^2$ and $\ell > 1$. □

Proof (The proof of Lemma 3.2). The proof follows from the results in [42], where McCoy defined the generalized notion of the rank of matrices over a commutative ring R and showed that a homogeneous equation has a non-trivial solution over R if and only if the rank of the coefficient matrix is less than the number of unknowns.

Assume $E = \{\alpha_1, \alpha_2, \ldots, \alpha_\ell\}$. Then the homogeneous linear Eqs. (10) can be written as

$$\begin{pmatrix} w(\alpha_1,z) & w(\alpha_1,z)\alpha_1 & \ldots & w(\alpha_1,z)\alpha_1^e & -1 & -\alpha_1 & \ldots & -\alpha_1^{\rho\ell-1+e} \\ w(\alpha_2,z) & w(\alpha_2,z)\alpha_2 & \ldots & w(\alpha_2,z)\alpha_2^e & -1 & -\alpha_2 & \ldots & -\alpha_2^{\rho\ell-1+e} \\ \vdots & \vdots & \ldots & \vdots & \vdots & \vdots & \ldots & \vdots \\ w(\alpha_\ell,z) & w(\alpha_\ell,z)\alpha_\ell & \ldots & w(\alpha_\ell,z)\alpha_\ell^e & -1 & -\alpha_\ell & \ldots & -\alpha_\ell^{\rho\ell-1+e} \end{pmatrix} \begin{pmatrix} A_0(z) \\ A_1(z) \\ \vdots \\ A_e(z) \\ B_0(z) \\ B_1(z) \\ \vdots \\ B_{\rho\ell-1+e}(z) \end{pmatrix} = 0$$

Let $M(z)$ be the coefficient matrix. The rank of $M(z)$ [42] is defined as the greatest positive integer $r \leq 2e + \rho\ell + 1$ such that the set of all $r \times r$ subdeterminants of $M(x)$ does not have a common non-zero annihilator.

The above linear equations over $\mathsf{GR}[z]$ have $2e + \ell\rho + 1$ unknowns. For any $(2e+\rho\ell+1)\times(2e+\rho\ell+1)$ minor of $M(z)$, let $R(z)$ denote its determinant. Since the entries in $M(z)$ that contribute nonzero degrees with respect to z come from $\{w(\alpha_i, z) : i = 1, 2 \ldots, \ell\}$ and each $\deg_z(w(\alpha_i, z)) \leq s$, we have $R(z)$ has degree at most $s(e+1)$. However, by taking $z = \gamma$ for $\gamma \in S$, the above linear equations have nonzero solutions because $\mathfrak{d}(w(x,\gamma), \mathsf{RS}) \leq (1-\rho)/2$. Thus $R(\gamma) = 0$ for all $\gamma \in S$. Since $S \subset T$ and $|S| > s(e+1)$, we have that $R(z) = 0$ according to Lemma 2.1. Thus the rank of $M(z)$ is less than the number of unknowns. By [42, Theorem 1], Eq. (10) has a tuple of solutions $A(x,z), B(x,z) \in \mathsf{GR}[x,z]$ [42, Theorem 1]. Furthermore, assume r is the rank of $M(z)$ and without loss of generality, assume this minor is in the upper left-hand corner of $M(x)$. Let $d_1, d_2, \ldots, d_{r+1}$ be the respective co-factors of the first $r+1$ elements in the $r+1$-th row. Then a tuple of solutions of $M(z)\mathbf{x}^T = 0$ is given by

$$x_j = ad_j,\ j = 1, 2, \ldots, r+1; x_j = 0,\ j = r+2, \ldots, 2e+\rho\ell+1,$$

where $a \in \mathsf{GR}[z]$ is a nonzero element that annihilates the set of all determinants of square minors of $M(z)$ of order $r+1$. That means we can take $A_i(z), B_i(z) \in \{ad_1, ad_2, \ldots, ad_{r+1}, 0\} \subset \mathsf{GR}[z]$. Thus, there exist $A(x,z) = \sum_{i=0}^{e} A_i(z)x^i, B(x,z) = \sum_{i=0}^{e+\rho\ell-1} B_i(z)x^i \in \mathsf{GR}[x,z]$ satisfying Eq. (1). Note that if $A_i(z) = ad_j$ (resp. $B_i(z) = ad_j$) for some $j = 1, 2, \ldots, r+1$, then d_j has degree at most se (resp. $s(e+1)$). To complete the proof, it suffices to show that $a \in \mathsf{GR}$, which implies that

$$\deg_z(A) \leq se,\ \deg_z(B) \leq s(e+1).$$

Assume $f_i(z)$ is a nonzero determinant of some minor of $M(z)$ of order $r+1$. By the definition of rank, $f_i(z)$ is a divisor of zero in $\mathsf{GR}[z]$. Then by [42, Theorem

2], $f_i(z)$ is annihilated by a nonzero element a_i in GR. Assume $a_i = p^{k_i} a_{i,0}$ with $a_{i,0}$ is a unit in GR. Then p^{k_i} also annihilates f_i. Assume I is the index set of all nonzero determinant $f_i(z)$ of order $r+1$. Let

$$k_m = \max_{i \in I}\{\min\{k_i \mid p^{k_i} f_i(z) = 0\}\}.$$

Then $a = p^{k_m} \neq 0$ annihilates all $f_i(x)$ for $i \in I$. □

Proof (The proof of Lemma 3.3) Our proof recursively makes use of the Polishchuk–Spielman Lemma over the finite field $\mathbb{F}_{p^d}$ and the lifting techniques.

By Proposition 2.1, A, B, M_α, N_γ can be represented as $A = \sum_{i=0}^{k-1} A^{(i)}(x,z)p^i$, $B = \sum_{i=0}^{k-1} B^{(i)}(x,z)p^i$, $M_\alpha(z) = \sum_{i=0}^{k-1} M_\alpha^{(i)}(z)p^i$ and $N_\gamma(x) = \sum_{i=0}^{k-1} N_\gamma^{(i)}(x)p^i$, respectively, where $A^{(i)}(x,z)$ and $B^{(i)}(x,z)$ have degrees

$$\deg_x(A^{(i)}) \leq \deg_x(A) \leq a_x,\ \deg_x(B^{(i)}) \leq \deg_x(B) \leq b_x,$$
$$\deg_z(A^{(i)}) \leq \deg_z(A) \leq a_z,\ \deg_z(B^{(i)}) \leq \deg_z(B) \leq b_z.$$

For $0 \leq j \leq k-1$, let

$$A_j = A \bmod p^{j+1},\ B_j = B \bmod p^{j+1},\ M_{\alpha,j} = M_\alpha \bmod p^{j+1},\ N_{\gamma,j} = N_\gamma \bmod p^{j+1}.$$

Then $A_0(x,z), B_0(x,z), M_{\alpha,0}(z)$ and $N_{\gamma,0}(x)$ are polynomials in $\mathbb{F}_{p^d}[x,z]$. Since $T \bmod p = \mathbb{F}_{p^d}$, for any two elements $\alpha, \beta \in T$, $(\bar{\alpha} := \alpha \bmod p) = (\bar{\beta} := \beta \bmod p)$ if and only if $\alpha = \beta$. Thus, there are still n_x elements $\bar{\alpha}$ in $\mathbb{F}_{p^d}$ such that

$$A_0(\alpha, z) M_{\alpha,0}(x) = B_0(\alpha, z) \bmod p. \tag{13}$$

Also, there are still n_z elements $\bar{\gamma}$ in $\mathbb{F}_{p^d}$ such that

$$A_0(x, \gamma) N_{\gamma,0}(x) = B_0(x, \gamma) \bmod p. \tag{14}$$

Now that all the conditions in the Polishchuk–Spielman lemma [11, Lemma 4.3] over the finite field $\mathbb{F}_{p^d}$ are satisfied, there is a polynomial $P_0(x,z) \in \mathsf{GR}[x,z]$ with bounded degrees $\deg_x(P_0) \leq d_x,\ \deg_z(P_0) \leq d_z$ such that:

$$A_0 P_0 = B_0 \bmod p.$$

Let $T_x := \{\alpha \in T \mid A_0(\alpha, x) = 0\}$ and $T_z := \{\gamma \in T \mid A_0(x, \gamma) = 0\}$. Then from Eq. (13) and (14), we have $P_0(\alpha, z) = M_{\alpha,0}(z)$ for at least $n_x - a_x$ $\alpha \in \backslash T_x$, and $P_0(x, \gamma) = N_{\gamma,0}(x)$ for at least $n_z - a_z$ $\gamma \in T \setminus T_z$, respectively.

Next, we show the existence of $P(x,z) = \sum_{i=0}^{k-1} P^{(i)} p^i$ by induction on i. Suppose that, for $j \geq 0$ we have a polynomial $P_j = \sum_{i \leq j} P^{(i)} p^i$ with bounded degrees $\deg_x(P_j) \leq d_x,\ \deg_z(P_j) \leq d_z$ satisfying

1) $A_j P_j = B_j \bmod p^{j+1}$;
2) $P_j(\alpha, z) = M_{\alpha,j}(z)$ for at least $n_x - a_x$ $\alpha \in E \setminus T_x$;
3) $P_j(x, \gamma) = N_{\gamma,j}(x)$ for at least $n_z - a_z$ $\gamma \in T \setminus T_z$.

According to $A_{j+1}(\alpha, z)M_{\alpha,j+1}(z) = B_{j+1}(\alpha, z) \bmod p^{j+2}$, we have

$$A_j(\alpha, z)M_{\alpha,j+1}(z) = \frac{B_j(\alpha, z) - A_j M_{\alpha,j}(z)}{p^{j+1}} + B^{(j+1)}(\alpha, z) - A^{(j+1)}M_{\alpha,j}(z) \mod p.$$

From assumption 2), we substitute the last $M_{\alpha,j}(z)$ in the right hand by $P_j(\alpha, z)$ and add a term $\frac{A_j(M_{\alpha,j}(z)-P_j(\alpha,z))}{p^{j+1}}$ to both sides. Then, for at least $n_x - a_x$ $\alpha \in E$, we have

$$\begin{aligned} &A_j(\alpha, z)\left(M_{\alpha,j+1}(z) + \frac{M_{\alpha,j}(z) - P_j(\alpha, z)}{p^{j+1}}\right) \\ &= \frac{B_j(\alpha, z) - A_j(\alpha, z)P_j(\alpha, z)}{p^{j+1}} + B^{(j+1)}(\alpha, z) - A^{(j+1)}P_j(\alpha, z) \mod p. \end{aligned}$$

Similarly, by the same analysis, for at least $n_z - a_z$ $\gamma \in T$, we have

$$\begin{aligned} &A_0(x, \gamma)\left(N_{\gamma,j+1}(x) + \frac{N_{\gamma,j} - P_j(x, \gamma)}{p^{j+1}}\right) \\ &= \frac{(B_j(x, \gamma) - A_j P_j(x, \gamma))}{p^{j+1}} + B^{(j+1)}(x, \gamma) - A^{(j+1)}P_j(x, \gamma) \mod p. \end{aligned}$$

Again, by the Polishchuk–Spielman lemma over $\mathbb{F}_{p^d}$, if $\frac{b_x}{n_x - a_x} + \frac{b_z}{n_z - a_z} < 1$, then there exists a polynomial $P^{(j+1)} \in \mathbb{F}_{p^d}[x, z]$ of bounded degrees $\deg_x(P^{(j+1)}) \leq d_x$, $\deg_z(P^{(j+1)}) \leq d_z$ such that

$$A_j P^{(j+1)} = \frac{B_j - A_j P_j}{p^{j+1}} + B^{(j+1)}(\alpha, z) - A^{(j+1)}P_j(x, \gamma) \mod p.$$

Let $P_{j+1} = P_j + p^{j+1}P^{(j+1)}$. Then $\deg_x(P_{(j+1)}) \leq d_x$, $\deg_z(P_{(j+1)}) \leq d_z$ and

$$A_{j+1}P_{j+1} = B_{j+1} \bmod p^{j+2}.$$

Moreover, for those γ defined as in 3), $P_{j+1}(x, \gamma) = N_{\gamma,j+1}(x)$, and for those α defined as in 2), $P_{j+1}(\alpha, z) = M_{\alpha,j+1}(z)$. Thus all assumptions 1)-3) still hold for P_{j+1}. By induction, there is a polynomial $P(x, z) = P_{k-1} = \sum_{i=0}^{k-1} P^{(i)}(x, z)p^i \in \mathsf{GR}$ satisfying 1)-3). We have proved. □

4 Polynomial Commitments from Reed-Solomon Codes

4.1 FRI-Based Polynomial Commitment

In this section, we show how to construct polynomial commitments for $\mathsf{GR}(2^k, d)$ by generalizing the FRI [10] approach.

Polynomial Commitments from Reed-Solomon IOPP. Let $L \subset \mathsf{GR}(2^k, d)$ be an exceptional set with $|L| = \ell$, and $\rho \in (0, 1)$. A Reed-Solomon IOP of Proximity allows a prover $\mathcal{P}$ to convince a verifier $\mathcal{V}$ a vector w is close to a Reed-Solomon code $\mathsf{RS}[L, \rho]$, where $\mathcal{V}$ has oracle access to w. The celebrated

work FRI [10] gives an efficient IOPP with linear prover computation, and logarithmic query complexity and verifier computation. Since an RS codeword f uniquely defines a polynomial $\hat{f}$ of degree less then $\rho\ell$, constructing polynomial commitment schemes from FRI follows a standard reduction. Specifically, $\mathcal{P}$ first sends to $\mathcal{V}$ an oracle message $f \in \mathsf{GR}^\ell$ claimed to be in $\mathsf{RS}[L, \rho]$, who then in the evaluation phase, picks a random point α and asks $\mathcal{P}$ to reveal $v := \hat{f}(\alpha)$, and the prover $\mathcal{P}$ is required to convince $\mathcal{V}$ that v is the correct evaluation. Observe that $\mathcal{V}$ has oracle access to a "virtual" oracle $g \in \mathsf{RS}[L, \rho']$, where $\hat{g} := \frac{\hat{f}(X)-v}{X-\alpha}$ and $\rho' = \rho - 1/\ell$, in the sense that query to g at some $\beta \in L$ can be emulated by querying to f at β, and computes $g(\beta) := \frac{f(\beta)-v}{\beta-\alpha}$. Then it suffices to test whether g is close to $\mathsf{RS}[L, \rho']$, and the intuition is that if $\hat{f}$ is of higher degree than $\rho\ell$ or $v \neq \hat{f}(\alpha)$, then g is far from $\mathsf{RS}[L, \rho']$. However, the above approach has an issue in computing $g(\beta)$, since $\beta - \alpha$ is not always invertible for all $\alpha \in \mathsf{GR}$. To this end, by Lemma 2.2, we can restrict $\alpha \in T$ and let $L \subset T$, where $T := \{0, 1, \ldots, \zeta^{2^d-2}\}$, and $\zeta \in \mathsf{GR}(2^k, d)$ has order $2^d - 1$. We remark that such PCS with queries restricted on T are sufficient for our polynomial IOPs in Sect. 5.

Fast Reed-Solomon IOP of Proximity for Galois Rings. We briefly review FRI for fields, where the core is a "folding" technique. Informally, let s be the folding parameter and β be a point, there exists a folding map $\mathsf{Fold} : \mathsf{RS}[L, \rho] \times \mathbb{F} \to \mathsf{RS}[L', \rho], (f, \beta) \mapsto f_\beta := \mathsf{Fold}(f, \beta)$, where $L', L \subset \mathbb{F}$ with $|L'| = |L|/s$, such that:

1. If $f \in \mathsf{RS}[L, \rho]$, then for every $\beta \in \mathbb{F}$, $f_\beta \in \mathsf{RS}[L', \rho]$;
2. If f is δ-far from $\mathsf{RS}[L, \rho]$ for $\delta \in (0, 1 - \sqrt{\rho})$, then f_β is δ-far from $\mathsf{RS}[L', \rho]$, with probability at least $1 - \mathrm{poly}(|L|)/|\mathbb{F}|$ over the choice of β.
3. There exists an algorithm that given β, and oracle access to f, can output $f_\beta(\gamma)$ for any $\gamma \in \mathbb{F}$, by making s queries to f.

The folding map allows to reduce the proximity test problem for $f \in \mathsf{RS}[L, \rho]$ to $f_\beta \in \mathsf{RS}[L', \rho]$, in the sense that by property 3 the prover is forced to send the desired oracle message f_β, then by properties $1, 2$, $\mathcal{V}$ accepts f *iff* he accepts f_β. Such folding maps are known to exist only for certain settings depending on the field structure. For the "multiplicative" flavor of folding, the underlying field $\mathbb{F}$ is required to have a sufficiently large s-smooth multiplicative subgroup, where s is expected to be a small constant, e.g., taking $\mathbb{F} = \mathbb{F}_{257}$, then $|\mathbb{F}^*| = 2^8$ is 2-smooth. While for fields $\mathbb{F}_{p^d}$ with p a small prime, there is an "additive" flavor of folding, where L, L' can be instantiated with $\mathbb{F}_p$-linear subspaces of $\mathbb{F}_{p^d}$.

However, for Galois rings $\mathsf{GR}(2^k, d)$ with $k > 1$, the "additive" folding cannot be applied, as they lack sufficiently large additive subgroups that are exceptional. Note that RS code should be defined over an exceptional set of the Galois rings. Hence, we resort to the "multiplicative" folding, where the goal is to find $O(1)$-smooth multiplicative subgroups of Galois rings. Thus, we let $\zeta \in \mathsf{GR}(2^k, d)$ with order $2^d - 1$ and denote $T := \{0, 1, \ldots, \zeta^{2^d-2}\}$. Now, it suffices to find $O(1)$-smooth multiplicative subgroups of T^*. However, we can only expect T^* to have s-smoothness with s a prime >2, and T^* is not smooth at all for most

d. Therefore, the FRI-based approach can only work for certain choices of d. Fortunately, this is still useful for our polynomial IOPs over $\mathbb{Z}_{2^k}$ in Sect. 5.

To find an appropriate d that yields a smooth T^* with $|T^*| = 2^d - 1$, we apply Euler's theorem. Namely, denote by φ the Euler function, we have that $q \mid 2^{\varphi(q)} - 1$, which means that, for instance, we can take $q = 3^4$, and the corresponding Galois ring $\mathsf{GR}(2^k, 54)$ has a multiplicative subgroup of size 3^4, from which we can commit to polynomials of degree $< 3^4$. Now, suppose $T^* \subset \mathsf{GR}(2^k, d)$ is s-smooth, and $L \subset T^*$ has cardinality some power to s, we define the folding map Fold for Galois rings as follows:

Suppose $f \in \mathsf{RS}[L, \rho]$, and denote $\hat{f}$ as the polynomial over $\mathsf{GR}(2^k, d)$ of degree less than $\rho|L|$ that agrees with f on L. We can uniquely write

$$\hat{f}(x) = \hat{f}^{(0)}(x^s) + x \cdot \hat{f}^{(1)}(x^s) + \cdots + x^{s-1} \cdot \hat{f}^{(s-1)}(x^s),$$

then, $f_\beta \in \mathsf{RS}[L', \rho]$ is given by

$$\hat{f}_\beta(x) := \hat{f}^{(0)}(x) + \beta \cdot \hat{f}^{(1)}(x) + \cdots + \beta^{s-1} \cdot \hat{f}^{(s-1)}(x),$$

where $L' := \{a^s \mid a \in L\}$. It holds that for a random $\gamma \xleftarrow{\$} L'$, and $\hat{h}(x)$ satisfying $\hat{h}(\gamma_j) = f(\gamma_j)$, for each $j \in [1, s]$, where $\{\gamma_1, \ldots, \gamma_s\}$ are the roots of $x^s - \gamma$ in L, $\hat{h}(\beta) = f_\beta(\gamma)$. A self-contained PCS based on FRI is given in π_{FRICom} in the full version.

Theorem 4.1 *By the BCS compiler [14], π_{FRICom} yields a polynomial commitment scheme with $O(n)$ prover time, $O(\lambda \log^2 n)$ proof size and verifier time.*

Proof Correctness can be verified easily. The soundness analysis relies on the following claim: Let $f_i \in \mathsf{RS}[L, \rho]$, for $i \in [1, r]$, and suppose that at least one of f_i has relative distance at least δ from $\mathsf{RS}[L, \rho]$, then with high probability a random vector in the span of $f_1, \ldots, f_r$ also is at least δ far from $\mathsf{RS}[L, \rho]$. Actually, we prove this claim in Theorem 3.3. Then by [15, Corollary 5.5 & Theorem 5.11], the round-by-round knowledge soundness error of FRI is

$$\epsilon_{rbr}^{\mathrm{FRI}} = \max\{\frac{s\ell}{2^d}, (1-\delta)^m\},$$

where m is the repetition parameter, i.e., Step $3, 4, 5$ of π_{FRICom} would be parallel repeated for m times for guaranteeing security. The left proof about the prover's and verifier's time is very similar to that of [15], and it is omitted here. □

4.2 Ligero-Based Polynomial Commitment

In this section, we show how to construct polynomial commitments for Galois rings by generalizing the Ligero [3] approach. Let $\mathsf{GR}(2^k, d)$ be a Galois ring, and $f(X) \in \mathsf{GR}[X]$ be a polynomial of degree less than n. Intuitively, the construction is based on a fact that the evaluation of f on a point r is equivalent to the inner product $\langle \boldsymbol{f}, \boldsymbol{r} \rangle$, where $\boldsymbol{f} = (f_0, f_1, \ldots, f_{n-1})$ specifies the coefficients of f and

$\boldsymbol{r} = (1, r, r^2, \ldots, r^{n-1})$. Moreover, the inner product involves a tensor-product structure in the following sense. W.o.l.g., we assume $n = m^2$ for some integer m and define $\boldsymbol{a} := (1, r, r^2, \ldots, r^{m-1})$ and $\boldsymbol{b} := (1, r^m, r^{2m}, \ldots, r^{(m-1)m})$. In addition, $\boldsymbol{f}$ can be viewed as an $m \times m$ matrix with entries indexed as $(f_{0,0}, \ldots, f_{m-1,m-1})$. Namely, we have that $f_{i,j} = f_{i \cdot m+j}$, where $i, j \in [0, m-1]$. Then we have that

$$\begin{aligned}\langle \boldsymbol{f}, \boldsymbol{r} \rangle &= \sum_{i=0}^{n-1} f_i \cdot r^i = \sum_{i=0}^{m-1} \sum_{j=0}^{m-1} f_{i \cdot m+j} \cdot r^{i \cdot m+j} \\ &= \sum_{i=0}^{m-1} (r^m)^i \cdot \sum_{j=0}^{m-1} f_{i,j} \cdot r^j = \boldsymbol{b}^{\mathrm{T}} \cdot (f_{i,j})_{i,j \in [0,m-1]} \cdot \boldsymbol{a}.\end{aligned}$$

Let $\boldsymbol{v} \in \mathsf{GR}^m$ denote the row vector $\boldsymbol{b}^{\mathrm{T}} \cdot (f_{i,j})_{i,j \in [0,m-1]}$, which is actually a row-span of $(f_{i,j})_{i,j \in [0,m-1]}$. Hence, it suffices for $\mathcal{V}$ to verify the correctness of $\boldsymbol{v}$, from which $\mathcal{V}$ can compute $f(r) = \boldsymbol{v} \cdot \boldsymbol{a}$ locally.

Let enc be an encoding algorithm of any linear code and denote each row of $(f_{i,j})_{i,j \in [0,m-1]}$ by $f^{(i)} := (f_{i,0}, \ldots, f_{i,m-1})$, for $i \in [0, m-1]$. The core observation is that by the linearity:

$$\begin{aligned}\mathsf{enc}(\boldsymbol{v}) = \mathsf{enc}(\boldsymbol{b}^{\mathrm{T}} \cdot (f_{i,j})_{i,j \in [0,m-1]}) &= \mathsf{enc}(\sum_{i=0}^{m-1} b_i \cdot f^{(i)}) \\ &= \sum_{i=0}^{m-1} b_i \cdot \mathsf{enc}(f^{(i)}) = \sum_{i=0}^{m-1} b_i \cdot G^{(i)},\end{aligned}$$

where $G^{(i)} = \mathsf{enc}(f^{(i)})$. The high-level idea is that if $\boldsymbol{v}^*$ sent by $\mathcal{P}$ differs from $\boldsymbol{v}$, then the encodings of $\boldsymbol{v}^*, \boldsymbol{v}$ differ at least D entries, where D is the minimal distance of the underlying code. Therefore, $\mathcal{V}$ can check by looking at a random subset J of entries, and rejects if $\mathsf{enc}(\boldsymbol{v})|_j \neq \mathsf{enc}(\boldsymbol{v}^*)|_j$, for some $j \in J$.

Specifically, we use a Reed-Solomon code $\mathsf{RS}[E, \rho]$, where E is an exceptional set of GR and $\rho = m/|E| = m/\ell$, and view $G := (G^{(0)}, \ldots, G^{(m-1)})^{\intercal}$ as a codeword of the interleaved RS code RS^m. Suppose $\mathcal{V}$ is given $G \in \mathsf{GR}^{m \times \ell}$ in the clear, then $\mathcal{V}$ can check whether $G \in \mathsf{RS}^m$ and obtain $\mathsf{enc}(\boldsymbol{v})|_J := (\boldsymbol{b}^{\mathrm{T}} G)|_J$. Essentially, $\mathcal{V}$ does not need to read G entirely, and moreover, it suffices to convince $\mathcal{V}$ that G is close to RS^m. Therefore, our Ligero-based polynomial commitment scheme has the following structure: in the commit phase, the prover $\mathcal{P}$ sends oracle message G to $\mathcal{V}$ and claims that G is close to RS^m through an IOPP for interleaved RS codes; then in the evaluation phase, $\mathcal{V}$ selects a random r and sends $\boldsymbol{b}$ to $\mathcal{P}$, who then returns $\boldsymbol{v}$ and $\mathcal{V}$ can check the validity of $\boldsymbol{v}$ by querying columns of G.

Now it suffices to show the IOPP for RS^m, which follows the intuition that if G is far from RS^m, then the random row-span of G is still far from RS with high probability. This indicates that $\mathcal{V}$ can check by asking $\mathcal{P}$ to provide an RS codeword, which is claimed to be a random row-span of G, and comparing with that computed from querying columns of G. Here, we emphasis that our analyses show that the above intuition only holds for $\mathsf{GR}(2^k, d)$ that has a sufficiently large

extension degree d. We give a self-contained PCS construction in $\pi_{\mathsf{LigeroCom}}$ in the full version.

Complexity: Suppose GR is FFT-friendly, then prover has computation complexity of $O(n \log m)$, and verifier has computation complexity of $O(m \log m)$, with query complexity $2t$, oracle length $O(n)$.

Knowledge Soundness: By the $(1-\rho)/2$ proximity gap of Theorem 3.3, we have Theorem 4.2, whose detailed proofs are given in the full version. In particular, proof of knowledge can also be obtained when applying the BCS compiler [14] to $\pi_{\mathsf{LigeroCom}}$. To show the existence of an extractor, it suffices to show that a randomly chosen $N \times N$ matrix is invertible with high probability, which also holds for Galois rings $\mathsf{GR}(2^k, d)$ because $\mathsf{GR}(2^k, d)/(2) \simeq \mathbb{F}_{2^d}$. We refer to [54, Appendix C.4] for a formal proof.

Theorem 4.2. *Let D be the minimal distance of* RS*, and e be a positive integer such that $e < D/2$, then the binding soundness of $\pi_{\mathsf{LigeroCom}}$ is upper bounded by*

$$(1-\frac{e}{\ell})^t + \frac{\ell}{2^d} + (\frac{e+m}{\ell})^t.$$

5 IOP for $\mathbb{Z}_{2^k}$ from Degree-4 RMFE

In this section, we construct an IOP for proving R1CS relations over $\mathbb{Z}_{2^k}$ (see Π_{R1CS}). At a high-level, our results are obtained by incorporating Aurora [13] and degree-4 reverse-multiplication friendly embedding (RMFE) [28]. The intuitive idea is to use RMFEs to convert R1CS relations over $\mathbb{Z}_{2^k}$ to related relations over a Galois ring $\mathsf{GR}(2^k, d)$, and designing IOPs for such Galois ring relations can be adapted from existing IOPs for fields.

5.1 Characterizing R1CS Over $\mathbb{Z}_{p^k}$

Definition 5.1 (R1CS relation). *The relation $\mathbb{R}_{\mathrm{R1CS}}$ is the set of all pairs $(\mathbb{x}, \mathbb{w}) = ((\mathcal{R}, A, B, C, n_0, n_1, n_2, x), w)$, where $A, B, C \in \mathcal{R}^{n_1 \times n_2}$, $x \in \mathcal{R}^{n_0}$, $w \in \mathcal{R}^{n_2 - n_0}$, $z := (x, w) \in \mathcal{R}^{n_2}$, and $Az * Bz = Cz$.*

Here $\mathcal{R}$ is considered as $\mathbb{Z}_{p^k}$. Let (ϕ, ψ) be an $(m, d; 4)$-RMFE over $\mathbb{Z}_{p^k}$. Denote $\{\phi(x) \mid x \in \mathbb{Z}_{p^k}\}$ as $\mathrm{Im}(\phi)$, and $\{x \in \mathsf{GR}(p^k, d) \mid \psi(x) = 0\}$ as $\mathrm{Ker}(\psi)$.

Theorem 5.1. *Let $A, B, C \in \mathbb{Z}_{p^k}^{n_1 \times mn_2}$, $x \in \mathbb{Z}_{p^k}^{mn_0}$. If $n_1 \geq n_2$, then there exist $w \in \mathbb{Z}_{p^k}^{m(n_2 - n_0)}$ such that $((\mathbb{Z}_{p^k}, A, B, C, mn_0, n_1, mn_2, x), w) \in \mathbb{R}_{\mathrm{R1CS}}$ if and only if there exist $\hat{w} \in \mathrm{Im}(\phi)^{n_2 - n_0}$ and $\eta \in \mathrm{Ker}(\psi)^{n_1}$ such that*

$$\hat{A}\hat{z} * \hat{B}\hat{z} = \hat{C}\hat{z} + \eta,$$

where $\hat{z} := (\phi(x), \hat{w}) \in \mathsf{GR}^{n_2}$, and $\hat{A} := \phi(A), \hat{B} := \phi(B), \hat{C} := \phi(C) \in \mathsf{GR}^{n_1 \times n_2}$.

The proof of Theorem 5.1 is given in the full version. Here, we remark that previous works [16,21] reduce R1CS relations over $\mathbb{F}_2$ to $\mathbb{F}_{2^k}$ via degree-2 RMFEs, which can be directly applied to the $\mathbb{Z}_{2^k}$ setting as well. However, we observe that R1CS is essentially a degree-4 relation (of inputs A, B, C, x) hence it is much more compatible with degree-4 RMFEs. As a consequence, our transformation is simpler, yielding fewer queries.

From now on, let GR denote a Galois ring $\mathsf{GR}(p^k, d)$. Suppose $\zeta \in \mathsf{GR}$ has order $p^d - 1$ and let $T = \{0, 1, \zeta, \ldots, \zeta^{p^d-2}\}$. By Theorem 5.1, it suffices to consider following relations.

Definition 5.2 (Lincheck relation). *The relation $\mathbb{R}_{\mathrm{lin}}$ is the set of all pairs $(\mathbb{x}, \mathbb{w}) = \big((\mathsf{GR}, H_1, H_2, M), (f_1, f_2)\big)$, where H_1, H_2 are multiplicative subgroups of T^*, $f_1, f_2 \in \mathsf{GR}[X]$ have degree less than $|H_1|, |H_2|$, respectively, $M \in \mathsf{GR}^{H_1 \times H_2}$, and $\forall a \in H_1$, $f_1(a) = \sum_{b \in H_2} M_{a,b} \cdot f_2(b)$*[5].

Definition 5.3 (Rowcheck relation). *The relation $\mathbb{R}_{\mathrm{row}}$ is the set of all pairs $(\mathbb{x}, \mathbb{w}) = \big((\mathsf{GR}, H), (f_1, f_2, f_3)\big)$, where H is a multiplicative subgroup of T^*, $f_1, f_2, f_3 \in \mathsf{GR}[X]$ have degree less than $|H|$, and $\forall a \in H$, $f_1(a) \cdot f_2(a) = f_3(a)$.*

Definition 5.4 (Imcheck relation). *The relation $\mathbb{R}_{\mathrm{im}}$ is the set of all pairs $(\mathbb{x}, \mathbb{w}) = \big((\mathsf{GR}, H, \phi), f\big)$, where H is a multiplicative subgroup of T^*, $\phi : \mathbb{Z}_{p^k}^m \to \mathsf{GR}(p^k, d)$ is a $\mathbb{Z}_{p^k}$-linear map ($d > m$), $f \in \mathsf{GR}[X]$ has degree less than $|H|$, and $\forall a \in H$, $f(a) \in \mathrm{Im}(\phi)$.*

Definition 5.5 (Kercheck relation). *The relation $\mathbb{R}_{\mathrm{ker}}$ is the set of all pairs $(\mathbb{x}, \mathbb{w}) = \big((\mathsf{GR}, H, \psi), f\big)$, where H is a multiplicative subgroup of T^*, $\psi : \mathsf{GR}(p^k, d) \to \mathbb{Z}_{p^k}^m$ is a $\mathbb{Z}_{p^k}$-linear map ($d > m$), $f \in \mathsf{GR}[X]$ has degree less than $|H|$, and $\forall a \in H$, $f(a) \in \mathrm{Ker}(\psi)$.*

Given definitions as above, we have the following corollary.

Corollary 5.1. *Proving an R1CS relation $\mathbb{R}_{\mathrm{R1CS}} = ((\mathbb{Z}_{p^k}, A, B, C, mn_0, n_1, mn_2, x), w)$ is equivalent to proving the following relations:*

$$\begin{aligned}
&\mathbb{R}_{\mathrm{row}} = \big((\mathsf{GR}, H_1), (f_{\hat{A}\hat{z}}, f_{\hat{B}\hat{z}}, f_{\hat{C}\hat{z}} + f_\eta)\big),\\
&\mathbb{R}_{\mathrm{lin}}^M = \big((\mathsf{GR}, H_1, H_2, \hat{M}), (f_{\hat{M}\hat{z}}, f_{\hat{z}})\big), \text{ for } M \in \{A, B, C\},\\
&\mathbb{R}_{\mathrm{ker}} = \big((\mathsf{GR}, H_1, \psi), f_\eta\big),\\
&\mathbb{R}_{\mathrm{im}} = \big((\mathsf{GR}, H_2, \phi), f_{\hat{z}}\big),
\end{aligned}$$

where $\hat{z} = (\hat{x}, \hat{w}), |H_1| = n_1, |H_2| = n_2$, and H_1, H_2 are multiplicative subgroups of an exceptional set T of GR. We remark that we can apply a standard approach of [13] to guarantee $f_{\hat{z}}(a) = \hat{x}_a$ for all $a \in H_0$, where $H_0 \subset H_2$, and $|H_0| = n_0$.

Hence, it suffices to construct sub-protocols for checking the above four relations.

[5] In this paper we abuse a bijection between H_1 and $[1, |H_1|]$, and $M_{a,b}$ refers to the entry of M respect to row index a and column index b.

5.2 Univariate-Sumcheck for Galois Rings

Jumping ahead, we show how to prove a univariate Sumcheck relation in this section, which serves as a building block for proving Lincheck relations. Let GR denote a Galois ring $\mathsf{GR}(p^k, d)$, and $T = \{0, 1, \zeta, \ldots, \zeta^{p^d-2}\}$ with $\zeta \in \mathsf{GR}$ of order $p^d - 1$. Sumcheck relations are defined as follows.

Definition 5.6 (Sumcheck relation). *The relation $\mathbb{R}_{\text{sum}}$ is the set of all pairs $(\mathbb{x}, \mathbb{w}) = ((\mathsf{GR}, H, \mu), f)$, where H is a multiplicative subgroup of T^*, $f \in \mathsf{GR}[X]$, and $\sum_{a \in H} f(a) = \mu$.*

We adapt from [13] the reduction of Lincheck to Sumcheck for fields to Galois rings GR. Firstly, we have the following Lemma. The proof is given in the full version.

Lemma 5.1. *Let H be a multiplicative coset of $T^* = \{1, \zeta, \ldots, \zeta^{p^d-2}\}$ and $n = |H|$. Let $f(x)$ be a univariate polynomial over GR of degree strictly less than n. Then $\sum_{a \in H} f(a) = f(0) \cdot n$.*

For $f \in \mathsf{GR}[X]$ with $\deg(f) < n$, by Lemma 5.1, it suffices to have $\mathcal{P}$ send to $\mathcal{V}$ an oracle message $g \in \mathsf{GR}[X]$ with degree $< n - 1$, where $g(X) := \frac{f(X) - \mu/n}{X}$, so that $\mathcal{V}$ can query a random point α and check $\alpha \cdot g(\alpha) + \mu/n \stackrel{?}{=} f(\alpha)$. While for f with degree $\geq n$, f can be uniquely written as $f(X) := Xg(X) + \mu/n + Z_H(X)h(X)$, where $\deg(h) \leq \deg(f) - n$, and $Z_H(X)$ is the vanishing polynomial on H. Hence, in this case, $\mathcal{P}$ can send two oracle messages $g \in \mathsf{GR}[X]_{<n-1}, h \in \mathsf{GR}[X]_{\leq \deg(f)-n}$ to $\mathcal{V}$, who then tests $f(X) \stackrel{?}{=} Xg(X) + \mu/n + Z_H(X)h(X)$ on a random point α. We remark that n should be invertible in GR, i.e. n is coprime to p, which always holds since n divides $|T^*| = p^d - 1$.

Procedure 1: π_{Sumcheck}

Let $\mathsf{GR}(p^k, d)$ be a Galois ring, and $\zeta \in \mathsf{GR}(p^k, d)$ with order $p^d - 1$. Let $T = \{0, 1, \zeta, \ldots, \zeta^{p^d-2}\}$, and $H \subset T^*$. Let $n = |H|$.
Goal: Given an oracle access to $f \in \mathsf{GR}[X]$, Sumcheck $\sum_{a \in H} f(a) = \mu$.

1. $\mathcal{P}$ computes the unique polynomials g and h such that $\deg(g) < n - 1$, $\deg(h) \leq \deg(f) - n$, and $f(X) \equiv Xg(X) + \frac{\mu}{n} + Z_H(X)h(X)$.
2. $\mathcal{P}$ sends two oracles $g \in \mathsf{GR}[X]_{<n-1}, h \in \mathsf{GR}[X]_{\leq \deg(f)-n}$ to $\mathcal{V}$.
3. $\mathcal{V}$ samples $\alpha \stackrel{\$}{\leftarrow} T$, and query f, g, h at α.
4. $\mathcal{V}$ accepts if and only if $\alpha g(\alpha) = f(\alpha) - \frac{\mu}{n} - Z_H(\alpha)h(\alpha)$.

Soundness. Consider $f \in \mathsf{GR}[X]$ with $\sum_{a \in H} f(a) = \mu' \neq \mu$. Then for any $g \in \mathsf{GR}[X]_{<n-1}$, $h \in \mathsf{GR}[X]_{\leq \deg(f)-n}$ chosen by corrupted $\mathcal{P}$, $\mathcal{V}$ accepts if and only if α is a root of $f(X) - Xg(X) - \mu/n - Z_H h(X)$. Since $\alpha \stackrel{\$}{\leftarrow} T$, by Lemma 2.2, $\mathcal{V}$ accepts with probability at most $\frac{\max(\deg(f), n)}{|T|}$.
Efficiency. $\mathcal{V}$' running time is dominated by evaluating $Z_H(X)$ at some point α.

5.3 Rowcheck, Lincheck, Imcheck, and Kercheck

Recall that the goal of Rowcheck is to verify $f_1(a) \cdot f_2(a) = f_3(a)$ for all $a \in H$, given three polynomials $f_1, f_2, f_3 \in \mathsf{GR}[X]$ of degree less than $n = |H|$. Through defining $f := f_1 \cdot f_2 - f_3$, it is equivalent to check whether f vanishes on H, so as to check $Z_H(X)|f(X)$, where $Z_H(X)$ is the vanishing polynomial on H. Hence, it suffices to show the existence of $g \in \mathsf{GR}[X]$ of degree less than $n-1$, such that

$$g(X) \cdot Z_H(X) = f_1(X) \cdot f_2(X) - f_3(X). \tag{15}$$

To this end, we let $\mathcal{P}$ additionally send an oracle message $g \in \mathsf{GR}[X]_{<n-1}$ to $\mathcal{V}$, who then check Eq. (15) by querying g, f_1, f_2, f_3 at a random point. The construction is referred as π_{Rowcheck}.

Soundness. By Eq. (15), we know the malicious prover can select a non-zero polynomial $h^*(X) := g(X) \cdot Z_H(X) - f_1(X) \cdot f_2(X) + f_3(X)$ of degree $\leq 2n-2$ and wins if $h^*(\alpha) = 0$, where $\alpha \xleftarrow{\$} T$. Hence, by Lemma 2.2, $\mathcal{V}$ accepts with probability at most $\frac{2n-2}{|T|}$.

Efficiency. $\mathcal{V}$' running time is dominated by evaluating $Z_H(X)$ at some point α.

Procedure 2: π_{Rowcheck}

Let $\mathsf{GR}(p^k, d)$ be a Galois ring, and $\zeta \in \mathsf{GR}$ with order $p^d - 1$. Let $T = \{0, 1, \zeta, \ldots, \zeta^{p^d-2}\}$, and $H \subset T^*$. Let $\ell = |L|$, $n = |H|$. Denote by $Z_H(X)$ the vanish polynomial on H, i.e., $Z_H(X) := \prod_{a \in H}(X - a)$.

Goal: Given oracle access to $f_1, f_2, f_3 \in \mathsf{GR}[X]_{<n}$, check $f_1(a) \cdot f_2(a) = f_3(a)$ for all $a \in H$.

1. $\mathcal{P}$ defines a $g \in \mathsf{GR}[X]_{<n-1}$, such that $g(X) \cdot Z_H(X) = f_1(X) \cdot f_2(X) - f_3(X)$. $\mathcal{P}$ sends an oracle message g to $\mathcal{V}$.
2. $\mathcal{V}$ queries g, f_1, f_2, f_3 at a random point $\alpha \xleftarrow{\$} T$, and checks $g(\alpha) \cdot Z_H(\alpha) \stackrel{?}{=} f_1(\alpha) \cdot f_2(\alpha) - f_3(\alpha)$. $\mathcal{V}$ accepts if the check passes.

Recall in the Lincheck relation, $\mathcal{V}$ wants to check that $f_1(a) \stackrel{?}{=} \sum_{b \in H_2} M_{a,b} \cdot f_2(b)$ for all $a \in H_1$, given $M \in \mathsf{GR}^{n_1 \times n_2}$, and oracle access to $f_1 \in \mathsf{GR}[X]_{<n_1}, f_2 \in \mathsf{GR}[X]_{<n_2}$. By fixing a pre-defined order of elements in H_1, H_2, we write $\vec{f_1} \in \mathsf{GR}^{n_1}$ as the evaluations of f_1 on H_1 (similar for f_2), and the Lincheck relation is equivalent to check $\vec{f_1} \stackrel{?}{=} M \cdot \vec{f_2}$. To this end, we take a random linear combination on the entries, and it suffices to check the inner products $\left\langle \vec{f_1}, \vec{\gamma} \right\rangle \stackrel{?}{=} \left\langle M \cdot \vec{f_2}, \vec{\gamma} \right\rangle = \left\langle \vec{f_2}, M^\top \vec{\gamma} \right\rangle$ where $\vec{\gamma} \xleftarrow{\$} \mathsf{GR}^{n_1}$, which incurs a soundness error of $1/p^d$ by Lemma 2.2. This implies two Sumcheck relations that $v_1 = \sum_{a \in H_1}(f_1 \cdot f_{\vec{\gamma}})(a)$, $v_2 = \sum_{a \in H_2}(f_2 \cdot f_{M^\top \vec{\gamma}})(a)$, where $f_{\vec{\gamma}}, f_{M^\top \vec{\gamma}}$ are the extensions of $\vec{\gamma}, M^\top \vec{\gamma}$ on H_1, H_2, respectively, and v_1, v_2 should be equal. To reduce the invocations of Sumcheck, we let $H_2 \subseteq H_1$, and define $f_{M^\top \vec{\gamma}}$ such that $f_{M^\top \vec{\gamma}}(a) = 0$ for $a \in H_1 \backslash H_2$, and $f_{M^\top \vec{\gamma}}(a) = (M^\top \vec{\gamma})(a)$ for $a \in H_2$, if $n_1 \geq n_2$.

Hence, it suffices to Sumcheck $0 \stackrel{?}{=} \sum_{a \in H_1}(f_1 \cdot f_{\vec{\gamma}} - f_2 \cdot f_{M^\top \vec{\gamma}})(a)$. The case of $n_1 < n_2$ can be proceeded similarly. The construction is referred as π_{Lincheck}.

Soundness. From above discussions, the soundness of Lincheck consists two parts: $1/p^d$ from random linear combination, and that inherited from Sumcheck. We remark that $\mathcal{V}$ can only sample one $\gamma \stackrel{\$}{\leftarrow} \mathsf{GR}$ and set $\vec{\gamma} := (1, \gamma, \ldots, \gamma^{n_1-1})$, which slightly increases the soundness by n_1/p^d.

Efficiency. $\mathcal{V}$ needs to interpolate and evaluate $g_{M\vec{\gamma}}$ and $g_{\vec{\gamma}}$, due to emulating oracle access to $F(X) := (g_{\vec{\gamma}} f_1 - g_{M\vec{\gamma}} f_2)(X)$ from oracle access to $f_1(X), f_2(X)$. This dominates the computation of $\mathcal{V}$, in addition to that of running $\mathcal{V}_{\text{sum}}$.

Procedure 3: π_{Lincheck}

Let $\mathsf{GR}(p^k, d)$ be a Galois ring, and $\zeta \in \mathsf{GR}(p^k, d)$ with order $p^d - 1$. Let $M \in \mathsf{GR}(p^k, d)^{n_1 \times n_2}$, and $T = \{0, 1, \zeta, \ldots, \zeta^{p^d-2}\}$. If $n_1 \geq n_2$, let $H_2 \subseteq H_1 \subset T^*$, otherwise let $H_1 \subset H_2 \subset T^*$. Let $n_1 = |H_1|$, $n_2 = |H_2|$.

Goal: Given oracle access to $f_1 \in \mathsf{GR}[X]_{<n_1}$, $f_2 \in \mathsf{GR}[X]_{<n_2}$, Lincheck $f_1(a) = \sum_{b \in H_2} M_{a,b} \cdot f_2(b)$ for all $a \in H_1$.

1. $\mathcal{V}$ samples a random $\boldsymbol{\gamma} \stackrel{\$}{\leftarrow} \mathsf{GR}(p^k, d)^{n_1}$, and sends it to $\mathcal{P}$.
2. If $n_1 \geq n_2$, $\mathcal{P}$ and $\mathcal{V}$ interpolate polynomials g_γ and $g_{M\gamma}$ of degree less than n_1 such that for all $a \in H_1$, $g_\gamma(a) = \gamma_a$, and for $a \in H_2$, $g_{M\gamma}(a) = (M^\top \boldsymbol{\gamma})(a)$, for $a \in H_1 \backslash H_2$, $g_{M\gamma}(a) = 0$. Otherwise, $\mathcal{P}$ and $\mathcal{V}$ interpolate polynomials g_γ and $g_{M\gamma}$ of degree less than n_2 such that for all $a \in H_2$, $g_{M\gamma}(a) = (M^\top \boldsymbol{\gamma})(a)$, and for $a \in H_1$, $g_\gamma(a) = \gamma_a$, for $a \in H_2 \backslash H_1$, $g_\gamma(a) = 0$.
3. $\mathcal{P}$ can compute $F(X) := (g_\gamma f_1 - g_{M\gamma} f_2)(X)$, and $\mathcal{V}$ has oracle access to F[a].
4. If $n_1 \geq n_2$, $\mathcal{P}$ and $\mathcal{V}$ run a Sumcheck protocol $(\mathcal{P}_{\text{sum}}(\mathbb{x}, F), \mathcal{V}_{\text{sum}}(\mathbb{x}))$ with $\mathbb{x} = (\mathsf{GR}, H_1, 0)$. Otherwise, $\mathcal{P}$ and $\mathcal{V}$ run a Sumcheck protocol $(\mathcal{P}_{\text{sum}}(\mathbb{x}, F), \mathcal{V}_{\text{sum}}(\mathbb{x}))$ with $\mathbb{x} = (\mathsf{GR}, H_2, 0)$.
5. $\mathcal{V}$ accepts if and only if $\mathcal{V}_{\text{sum}}$ accepts.

[a] Query to F at some point $\alpha \in T$ can be emulated by querying f_1, f_2 at α, and computing $F(\alpha) := g_\gamma(\alpha) f_1(\alpha) - g_{M\gamma}(\alpha) f_2(\alpha)$.

Given a $\mathbb{Z}_{p^k}$-linear map $\phi : \mathbb{Z}_{p^k}^m \to \mathsf{GR}(p^k, d)$, the Imcheck checks whether $f(a) \in \mathrm{Im}(\phi)$ for all $a \in H$, where $f \in \mathsf{GR}[X]_{<n}$, and $n = |H|$. The intuitive idea is to check a random $\mathbb{Z}_{p^k}$-linear combination on $\vec{f}$. However, this gives only $1/p$ soundness, and hence needs to be repeated for several times to obtain a negligible soundness error. So the goal is to let $\mathcal{V}$ obtain an $r := \Gamma \cdot \vec{f} \in \mathsf{GR}^\lambda$, where $\Gamma \stackrel{\$}{\leftarrow} \mathbb{Z}_{p^k}^{\lambda \times n}$ is selected by $\mathcal{V}$, and λ is the security parameter. Though $\mathcal{V}$ has only oracle access to f, it suffices to delegate the computation of $\Gamma \cdot \vec{f}$ to $\mathcal{P}$ in a verifiable way. The main observation is that this can be reduced to a Lincheck relation, in which $\Gamma \cdot \vec{f}$ defines a polynomial f_r such that $f_r(a) = (\Gamma \cdot \vec{f})(a) =$

$\sum_{b\in H} \Gamma_{a,b} f(b)$, for $a \in H' \subset T$. Due to space constraints, π_{Imcheck} for Imcheck and π_{Kercheck} for Kercheck are given in the full version of this paper.

Soundness. From above discussions, the soundness of Imcheck (Kercheck) consists two parts: $1/p^\lambda$ from random linear combination, and that inherited from Lincheck.

Efficiency. $\mathcal{V}$ needs to interpolate and evaluate $f_r(X)$ himself, emulating $\mathcal{V}_{\text{lin}}$'s oracle access to $f_r(X)$. This dominates the computation of $\mathcal{V}$, in addition to that of running $\mathcal{V}_{\text{lin}}$.

Putting all Pieces Together: IOP for $\mathbb{Z}_{2^k}$. By Theorem 5.1, we combine π_{Lincheck}, π_{Rowcheck}, π_{Imcheck}, π_{Kercheck}, with degree-4 RMFEs, and give our IOP for R1CS over $\mathbb{Z}_{p^k}$ in Π_{R1CS}. Security and efficiency mainly follow [13] and are analyzed in the full version.

Protocol 4: Π_{R1CS}

Let $(\mathbb{x}, \mathbb{w}) = ((\mathbb{Z}_{p^k}, A, B, C, mn_0, n_1, mn_2, x), w)$ be an R1CS statement, $\mathsf{GR}(p^k, d)$ be a Galois ring, and $\zeta \in \mathsf{GR}(p^k, d)$ with order $p^d - 1$. Let $T = \{0, 1, \zeta, \ldots, \zeta^{p^d-2}\}$, and $H_0, H_1, H_2 \subset T$ with $|H_0| = n_0, |H_1| = n_1, |H_2| = n_2$, and $H_0 \subset H_2$. Let (ϕ, ψ) be an $(m, d; 4)$-RMFE over $\mathbb{Z}_{p^k}$.

1. $\mathcal{P}$ and $\mathcal{V}$ compute $\hat{M} := \phi(M) \in \mathsf{GR}^{n_1\times n_2}$, for $M \in \{A, B, C\}$, and $\hat{x} := \phi(x) \in \mathrm{Im}(\phi)^{n_0}$. In addition, $\mathcal{P}$ computes $\hat{w} := \phi(w) \in \mathrm{Im}(\phi)^{n_2-n_0}$, $\hat{z} := (\hat{x}, \hat{w})$ and $\eta := \hat{A}\hat{z} * \hat{B}\hat{z} - \hat{C}\hat{z} \in \mathrm{Ker}(\psi)^{n_1}$.
2. They compute $f_{\hat{x}}(X) \in \mathsf{GR}[X]_{<n_0}$ such that $f_{\hat{x}}(a) = \hat{x}_a$ for $a \in H_0$.
3. $\mathcal{P}$ computes $f_{\hat{w}} \in \mathsf{GR}[X]_{<n_2-n_0}$ such that $f_{\hat{w}}(a) = \frac{\hat{w}_a - f_{\hat{x}}(a)}{Z_{H_0}(a)}$[a], for $a \in H_2 \backslash H_0$. $\mathcal{P}$ also computes $f_{\hat{A}\hat{z}}, f_{\hat{B}\hat{z}}, f_{\hat{C}\hat{z}}, f_\eta \in \mathsf{GR}[X]_{<n_1}$ which agree with $\hat{A}\hat{z}, \hat{B}\hat{z}, \hat{C}\hat{z}, \eta$ on H_1, respectively.
4. $\mathcal{P}$ sends to $\mathcal{V}$ oracle messages $f_{\hat{A}\hat{z}}, f_{\hat{B}\hat{z}}, f_{\hat{C}\hat{z}}, f_\eta \in \mathsf{GR}[X]_{<n_1}$ and $f_{\hat{w}} \in \mathsf{GR}[X]_{<n_2-n_0}$. Essentially, $\mathcal{V}$ has an oracle access to $f_{\hat{z}}(X) := f_{\hat{w}}(X)Z_{H_0}(X) + f_{\hat{x}}(X)$, where $f_{\hat{z}} \in \mathsf{GR}[X]_{<n_2}$.
5. $\mathcal{P}$ and $\mathcal{V}$ run the following procedures in parallel:
   (a) A Rowcheck π_{Rowcheck} for $\mathbb{R}_{\text{row}} = \big((\mathsf{GR}, H_1), (f_{\hat{A}\hat{z}}, f_{\hat{B}\hat{z}}, f_{\hat{C}\hat{z}} + f_\eta)\big)$.
   (b) A Lincheck π_{Lincheck} for $\mathbb{R}_{\text{lin}} = \big((\mathsf{GR}, H_1, H_2, \hat{A}), (f_{\hat{A}\hat{z}}, f_{\hat{z}})\big)$.
   (c) A Lincheck π_{Lincheck} for $\mathbb{R}_{\text{lin}} = \big((\mathsf{GR}, H_1, H_2, \hat{B}), (f_{\hat{B}\hat{z}}, f_{\hat{z}})\big)$.
   (d) A Lincheck π_{Lincheck} for $\mathbb{R}_{\text{lin}} = \big((\mathsf{GR}, H_1, H_2, \hat{C}), (f_{\hat{C}\hat{z}}, f_{\hat{z}})\big)$.
   (e) A Kercheck protocol π_{Kercheck} for $\mathbb{R}_{\text{ker}} = \big((\mathsf{GR}, H_1, \psi), f_\eta\big)$.
   (f) An Imcheck protocol π_{Imcheck} for $\mathbb{R}_{\text{im}} = \big((\mathsf{GR}, H_2, \phi), f_{\hat{z}}\big)$.
6. $\mathcal{V}$ accepts if and only if all of the above subverifiers accept.

[a] Note that $Z_{H_0}(a) = \prod_{\alpha\in H_0}(a - \alpha)$ is invertible for any $a \in H_2 \backslash H_0$, since $H_2 \subset T$ is an exceptional set in GR.

6 Performance Evaluation

In this section, we evaluate the performance of our SNARKs for $\mathbb{Z}_{2^k}$. Specifically, we implement the following ingredients, i) Ligero-based PCS and FRI-based PCS for Galois rings, ii) Degree-D RMFEs for Galois rings, iii) IOP for $\mathbb{Z}_{2^k}$ from Degree-4 RMFEs. In particular, we also generalize the IOP for $\mathbb{F}_2$ from Degree-2 RMFEs in [21] to $\mathbb{Z}_{2^k}$ and implement it. We target on 128-bit security and focus on Galois rings $\mathsf{GR}(2^k, 162)$, since they contain 3-smooth multiplicative subgroups and support FFT algorithms. We use the C++ language and the NTL library for Galois ring operations. The benchmark performances are evaluated on a PC with Intel(R) Xeon(R) Gold 5220R 2.20 GHz CPU and 128 GB of RAM.

On the Computational Complexity. The prover time of our FRI-based SNARK mainly consists of three parts: (i) polynomial interpolation over Galois rings, (ii) FFT computation over Galois rings, and (iii) computation of Merkle-tree hashes. Among these, (i) and (ii) are highly influenced by the complexity of Galois ring arithmetic operation, and (ii) dominates the overall cost. Basically, our RMFE-based approaches improve the prover time by reducing the converted R1CS size over Galois rings. That is, with RMFEs, degrees of interpolated polynomials, as well as the FFT evaluation domain size, are reduced by a factor of $m = O(\lambda)$. However, operations over Galois rings are not as fast as those over finite fields and lack optimizations both algorithm-wise and implementation-wise.

Note that the inverse operations, such as inversions in Lagrange interpolation basis and small matrix inversions in FFT, can be precomputed. Therefore, it suffices to consider the multiplication operation over Galois rings. Since a single multiplication in $\mathsf{GR}(2^k, d)$ involves multiplying two polynomials in $\mathbb{Z}_{2^k}[x]$ of degree less than d, a naive implementation (as in the NTL library) requires $O(d^2)$ operations in $\mathbb{Z}_{2^k}$. On the one hand, this can be optimized to $O(d \log d \log \log d)$ using fast polynomial multiplication algorithms over general commutative rings [51, Theorem 8.23]. On the other hand, for some Galois rings of special structures, it is possible to achieve better efficiency. For instance, elements of $\mathsf{GR}(2^k, 162)$ can be viewed as polynomials over $\mathsf{GR}(2^k, 18)$, on which FFT can be applied. However, in our current implementation, we are not able to employ such optimizations due to the heavy use of the NTL library.

It turns out that on the NTL library interpolating a degree-16 polynomial over $\mathbb{F}_{2^{162}}$ already takes 16 ms! In comparison, it takes only 10.4 ms to interpolate a degree-4096 polynomial over $\mathbb{F}_{2^{192}}$ on the Libiop library. As a consequence, our implemented FRI-PCS and Ligero-PCS for the baseline case where $k = 1$ and $\mathsf{GR}(2^1, 162) = \mathbb{F}_{2^{162}}$ are much slower than highly optimized implementations in Brakedown [35]. However, given the above evidences, we are optimistic that there is much room for improvements on the concrete efficiency of our constructions. As past few years have seen the power of Galois rings in many other scenarios, we want to call for further studies on improving the concrete efficiency of performing Galois rings operations.

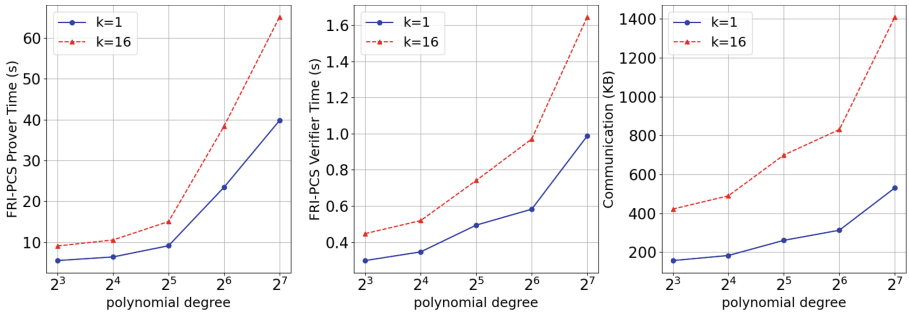

Fig. 1. Benchmark results of FRI-PCS on NTL.

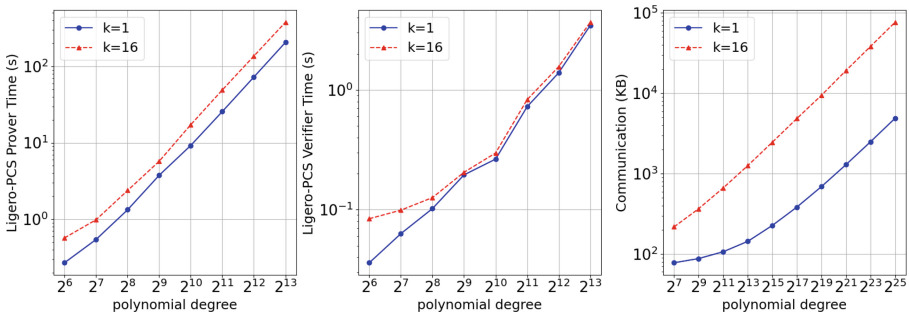

Fig. 2. Benchmark results of Ligero-PCS on NTL.

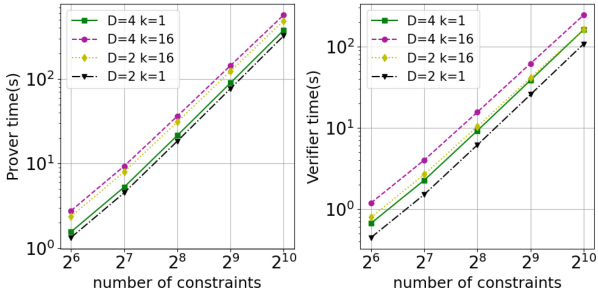

Fig. 3. Benchmark results of IOP for $\mathbb{Z}_{2^k}$ on NTL.

Benchmark of PCS for Galois Rings. Specifically, we evaluate the FRI-based PCS on degrees of $2^3, 2^4, 2^5, 2^6, 2^7$, and Ligero-based PCS on a wider range of polynomial degrees, from 2^6 to 2^{13}. We remark that the FRI-based PCS only supports polynomials of degree less than the maximum evaluation points set size, i.e., < 243 for $\mathsf{GR}(2^k, 162)$. We also calculate the proof size for both schemes. Due to the small evaluation point set, the numbers of our FRI-based PCS are

not promising. The results are summarized in Fig. 1 and Fig. 2, showing that the efficiency of $k = 16$ is comparable to that of $k = 1$.

Benchmark of IOP for $\mathbb{Z}_{2^k}$. Figure 3 reports the prover time of our IOP for $\mathbb{Z}_{2^k}$ from Degree-4 RMFEs and Degree-2 RMFEs, which is adapted from [21]. The underlying Galois ring is $\mathsf{GR}(2^k, 162)$, with $k = 1, 16$, where $(16, 162; 4)$-RMFEs and $(48, 162; 2)$-RMFEs over $\mathbb{Z}_{2^k}$ are employed, respectively. We select parameters for RMFEs according to [28], and the efficiency would be improved if Degree-4 RMFEs can be realized with better ratios. Generally speaking, Degree-2 RMFEs have better ratios than Degree-4 RMFEs, leading to R1CS over Galois rings with smaller size and less computation per FFT call. However, this comes at the cost of more FFT calls (though each is cheaper) and queries. Consequently, the Degree-2 RMFE-based approach yields faster proving but a larger argument size.

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References

1. Albrecht, M.R., Cini, V., Lai, R.W.F., Malavolta, G., Thyagarajan, S.A.K.: Lattice-based SNARKs: publicly verifiable, preprocessing, and recursively composable - (extended abstract). In: Dodis, Y., Shrimpton, T. (eds.) CRYPTO 2022. LNCS, vol. 13508, pp. 102–132. Springer, Cham (2022). https://doi.org/10.1007/978-3-031-15979-4_4
2. Albrecht, M.R., Fenzi, G., Lapiha, O., Nguyen, N.K.: SLAP: succinct lattice-based polynomial commitments from standard assumptions. In: Joye, M., Leander, G. (eds.) EUROCRYPT 2024. LNCS, vol. 14656, pp. 90–119. Springer, Cham (2024). https://doi.org/10.1007/978-3-031-58754-2_4
3. Ames, S., Hazay, C., Ishai, Y., Venkitasubramaniam, M.: Ligero: lightweight sublinear arguments without a trusted setup. In: CCS 2017, pp. 2087–2104. ACM (2017). https://doi.org/10.1145/3133956.3134104
4. Arnon, G., Chiesa, A., Fenzi, G., Yogev, E.: STIR: Reed-Solomon proximity testing with fewer queries. In: Reyzin, L., Stebila, D. (eds.) CRYPTO 2024. LNCS, vol. 14929, pp. 380–413. Springer, Cham (2024). https://doi.org/10.1007/978-3-031-68403-6_12
5. Arnon, G., Chiesa, A., Fenzi, G., Yogev, E.: WHIR: Reed-Solomon proximity testing with super-fast verification. In: Fehr, S., Fouque, P. (eds.) EUROCRYPT 2025. LNCS, vol. 15604, pp. 214–243. Springer, Cham (2025). https://doi.org/10.1007/978-3-031-91134-7_8
6. Arnon, G., Chiesa, A., Yogev, E.: IOPs with inverse polynomial soundness error. In: FOCS 2023, pp. 752–761. IEEE (2023). https://doi.org/10.1109/FOCS57990.2023.00050

7. Arun, A., Setty, S.T.V., Thaler, J.: Jolt: SNARKs for virtual machines via lookups. In: Joye, M., Leander, G. (eds.) EUROCRYPT 2024. LNCS, vol. 14656, pp. 3–33. Springer, Cham (2024). https://doi.org/10.1007/978-3-031-58751-1_1
8. Baum, C., Braun, L., Munch-Hansen, A., Razet, B., Scholl, P.: Appenzeller to brie: efficient zero-knowledge proofs for mixed-mode arithmetic and Z2k. In: CCS 2021, pp. 192–211. ACM (2021). https://doi.org/10.1145/3460120.3484812
9. Baum, C., Braun, L., Munch-Hansen, A., Scholl, P.: Moz$\mathbb{Z}_2^k$ arella: efficient vector-OLE and zero-knowledge proofs over $\mathbb{Z}_2^k$. In: Dodis, Y., Shrimpton, T. (eds.) CRYPTO 2022. LNCS, vol. 13510, pp. 329–358. Springer, Cham (2022). https://doi.org/10.1007/978-3-031-15985-5_12
10. Ben-Sasson, E., Bentov, I., Horesh, Y., Riabzev, M.: Fast Reed-Solomon interactive oracle proofs of proximity. In: ICALP 2018. LIPIcs, vol. 107, pp. 14:1–14:17. Schloss Dagstuhl - Leibniz-Zentrum für Informatik (2018). https://doi.org/10.4230/LIPIcs.ICALP.2018.14
11. Ben-Sasson, E., Carmon, D., Ishai, Y., Kopparty, S., Saraf, S.: Proximity gaps for Reed-Solomon codes. J. ACM **70**(5), 31:1–31:57 (2023). https://doi.org/10.1145/3614423
12. Ben-Sasson, E., Chiesa, A., Genkin, D., Tromer, E., Virza, M.: SNARKs for C: verifying program executions succinctly and in zero knowledge. In: Canetti, R., Garay, J.A. (eds.) CRYPTO 2013. LNCS, vol. 8043, pp. 90–108. Springer, Cham (2013). https://doi.org/10.1007/978-3-642-40084-1_6
13. Ben-Sasson, E., Chiesa, A., Riabzev, M., Spooner, N., Virza, M., Ward, N.P.: Aurora: transparent succinct arguments for R1CS. In: Ishai, Y., Rijmen, V. (eds.) EUROCRYPT 2019. LNCS, vol. 11476, pp. 103–128. Springer, Cham (2019). https://doi.org/10.1007/978-3-030-17653-2_4
14. Ben-Sasson, E., Chiesa, A., Spooner, N.: Interactive Oracle proofs. In: Hirt, M., Smith, A. (eds.) TCC 2016. LNCS, vol. 9986, pp. 31–60. Springer, Heidelberg (2016). https://doi.org/10.1007/978-3-662-53644-5_2
15. Block, A.R., Garreta, A., Katz, J., Thaler, J., Tiwari, P.R., Zając, M.: Fiat-shamir security of FRI and related SNARKs. In: Guo, J., Steinfeld, R. (eds.) ASIACRYPT 2023, vol. 14439, pp. 3–40. Springer, Singapore (2023)
16. Bootle, J., Chiesa, A., Guan, Z., Liu, S.: Linear-time probabilistic proofs with sublinear verification for algebraic automata over every field. IACR Cryptology ePrint Archive, p. 1056 (2022)
17. Braun, L., de Saint Guilhem, C.D., Jadoul, R., Orsini, E., Smart, N.P., Tanguy, T.: ZK-for-Z2K: MPC-in-the-head zero-knowledge proofs for $\mathbb{Z}_2^K$. In: Quaglia, E.A. (ed.) IMACC 2023. LNCS, vol. 14421, pp. 137–157. Springer, Cham (2023). https://doi.org/10.1007/978-3-031-47818-5_8
18. Brehm, M., Chen, B., Fisch, B., Resch, N., Rothblum, R.D., Zeilberger, H.: Blaze: fast SNARKs from interleaved RAA codes. In: Fehr, S., Fouque, P. (eds.) EUROCRYPT 2025. LNCS, vol. 15604, pp. 123–152. Springer, Cham (2025). https://doi.org/10.1007/978-3-031-91134-7_5
19. Bünz, B., Bootle, J., Boneh, D., Poelstra, A., Wuille, P., Maxwell, G.: Bulletproofs: short proofs for confidential transactions and more. In: SP 2018, pp. 315–334. IEEE Computer Society (2018). https://doi.org/10.1109/SP.2018.00020
20. Cascudo, I., Cramer, R., Xing, C., Yuan, C.: Amortized complexity of information-theoretically secure MPC revisited. In: Shacham, H., Boldyreva, A. (eds.) CRYPTO 2018. LNCS, vol. 10993, pp. 395–426. Springer, Cham (2018). https://doi.org/10.1007/978-3-319-96878-0_14

21. Cascudo, I., Giunta, E.: On interactive oracle proofs for Boolean R1CS statements. In: Eyal, I., Garay, J. (eds.) Financial Cryptography and Data Security. FC 2022. LNCS, vol. 13411, pp. 230–247. Springer, Cham (2022). https://doi.org/10.1007/978-3-031-18283-9_11
22. de Castro, L., Peikert, C.: Functional commitments for all functions, with transparent setup and from SIS. In: Hazay, C., Stam, M. (eds.) EUROCRYPT 2023. LNCS, vol. 14006, pp. 287–320. Springer, Cham (2023). https://doi.org/10.1007/978-3-031-30620-4_10
23. Chiesa, A., Hu, Y., Maller, M., Mishra, P., Vesely, P., Ward, N.P.: Marlin: preprocessing zkSNARKs with universal and updatable SRS. In: Canteaut, A., Ishai, Y. (eds.) EUROCRYPT 2020. LNCS, vol. 12105, pp. 738–768. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-45721-1_26
24. Cini, V., Malavolta, G., Nguyen, N.K., Wee, H.: Polynomial commitments from lattices: post-quantum security, fast verification and transparent setup. In: Reyzin, L., Stebila, D. (eds.) CRYPTO 2024. LNCS, vol. 14929, pp. 207–242. Springer, Cham (2024). https://doi.org/10.1007/978-3-031-68403-6_7
25. Cramer, R., Rambaud, M., Xing, C.: Asymptotically-good arithmetic secret sharing over $\mathbb{Z}/p^\ell\mathbb{Z}$ with strong multiplication and its applications to efficient MPC. In: Malkin, T., Peikert, C. (eds.) CRYPTO 2021. LNCS, vol. 12827, pp. 656–686. Springer, Cham (2021). https://doi.org/10.1007/978-3-030-84252-9_22
26. Dalskov, A.P.K., Escudero, D., Nof, A.: Fully secure MPC and zk-FLIOP over rings: new constructions, improvements and extensions. In: Reyzin, L., Stebila, D. (eds.) CRYPTO 2024. LNCS, vol. 14927, pp. 136–169. Springer, Cham (2024). https://doi.org/10.1007/978-3-031-68397-8_5
27. Diamond, B.E., Posen, J.: Succinct arguments over towers of binary fields. In: Fehr, S., Fouque, P. (eds.) EUROCRYPT 2025. LNCS, vol. 15604, pp. 93–122. Springer, Cham (2025). https://doi.org/10.1007/978-3-031-91134-7_4
28. Escudero, D., Hong, C., Liu, H., Xing, C., Yuan, C.: Degree-D reverse multiplication-friendly embeddings: constructions and applications. In: Guo, J., Steinfeld, R. (eds.) ASIACRYPT 2023. LNCS, vol. 14438, pp. 106–138. Springer, Singapore (2023). https://doi.org/10.1007/978-981-99-8721-4_4
29. Escudero, D., Song, Y., Wang, W.: Perfectly-secure multiparty computation with linear communication complexity over any modulus. In: Chung, K., Sasaki, Y. (eds.) ASIACRYPT 2024. LNCS, vol. 15489, pp. 102–132. Springer, Singapore (2024). https://doi.org/10.1007/978-981-96-0938-3_4
30. Escudero, D., Xing, C., Yuan, C.: More efficient dishonest majority secure computation over $\mathbb{Z}_2^k$ via Galois rings. In: Dodis, Y., Shrimpton, T. (eds.) CRYPTO 2022. LNCS, vol. 13507, pp. 383–412. Springer, Cham (2022). https://doi.org/10.1007/978-3-031-15802-5_14
31. Fisch, B., Liu, Z., Vesely, P.: Orbweaver: succinct linear functional commitments from lattices. In: Handschuh, H., Lysyanskaya, A. (eds.) CRYPTO 2023. LNCS, vol. 14082, pp. 106–131. Springer, Cham (2023). https://doi.org/10.1007/978-3-031-38545-2_4
32. Gabizon, A., Williamson, Z.J., Ciobotaru, O.: PLONK: permutations over Lagrange-Bases for oecumenical noninteractive arguments of knowledge. IACR Cryptology ePrint Archive, p. 953 (2019)
33. Ganesh, C., Nitulescu, A., Soria-Vazquez, E.: Rinocchio: SNARKs for ring arithmetic. J. Cryptol. **36**(4), 41 (2023). https://doi.org/10.1007/S00145-023-09481-3

34. Gao, Y., Kan, H., Li, Y.: Linear proximity gap for linear codes within the 1.5 Johnson bound. IACR Cryptology ePrint Archive, p. 1810 (2024). https://eprint.iacr.org/2024/1810
35. Golovnev, A., Lee, J., Setty, S.T.V., Thaler, J., Wahby, R.S.: Brakedown: linear-time and field-agnostic SNARKs for R1CS. In: Handschuh, H., Lysyanskaya, A. (eds.) CRYPTO 2023. LNCS, vol. 14082, pp. 193–226. Springer, Cham (2023). https://doi.org/10.1007/978-3-031-38545-2_7
36. Groth, J.: On the size of pairing-based non-interactive arguments. In: Fischlin, M., Coron, J.S. (eds.) EUROCRYPT 2016. LNCS, vol. 9666, pp. 305–326. Springer, Cham (2016). https://doi.org/10.1007/978-3-662-49896-5_11
37. Guruswami, V., Sudan, M.: Improved decoding of Reed-Solomon and algebraic-geometry codes. IEEE Trans. Inf. Theory **45**(6), 1757–1767 (1999). https://doi.org/10.1109/18.782097
38. Huang, M.M., Mao, X., Zhang, J.: Sublinear proofs over polynomial rings. IACR Cryptology ePrint Archive, p. 199 (2025). https://eprint.iacr.org/2025/199
39. Hwang, I., Seo, J., Song, Y.: Concretely efficient lattice-based polynomial commitment from standard assumptions. In: Reyzin, L., Stebila, D. (eds.) CRYPTO 2024. LNCS, vol. 14929, pp. 414–448. Springer, Cham (2024). https://doi.org/10.1007/978-3-031-68403-6_13
40. Lin, F., Xing, C., Yao, Y.: More efficient zero-knowledge protocols over $\mathbb{Z}_2^k$ via Galois rings. In: Reyzin, L., Stebila, D. (eds.) CRYPTO 2024. LNCS, vol. 14928, pp. 424–457. Springer, Cham (2024). https://doi.org/10.1007/978-3-031-68400-5_13
41. Lin, F., Xing, C., Yao, Y., Yuan, C.: Amortized NISC over $\mathbb{Z}_2^k$ from RMFE. In: Guo, J., Steinfeld, R. (eds.) ASIACRYPT 2023. LNCS, vol. 14438, pp. 38–70. Springer, Singapore (2023). https://doi.org/10.1007/978-981-99-8721-4_2
42. McCoy, N.H.: Remarks on divisors of zero. Am. Math. Mon. **49**(5), 286–295 (1942)
43. Nguyen, N.K., Seiler, G.: Greyhound: fast polynomial commitments from lattices. In: Reyzin, L., Stebila, D. (eds.) CRYPTO 2024. LNCS, vol. 14929, pp. 243–275. Springer, Cham (2024). https://doi.org/10.1007/978-3-031-68403-6_8
44. Parno, B., Howell, J., Gentry, C., Raykova, M.: Pinocchio: nearly practical verifiable computation. Commun. ACM **59**(2), 103–112 (2016). https://doi.org/10.1145/2856449
45. Polishchuk, A., Spielman, D.A.: Nearly-linear size holographic proofs. In: Leighton, F.T., Goodrich, M.T. (eds.) STOC 1994, pp. 194–203. ACM (1994). https://doi.org/10.1145/195058.195132
46. Quintin, G., Barbier, M., Chabot, C.: On generalized Reed-Solomon codes over commutative and noncommutative rings. IEEE Trans. Inf. Theory **59**(9), 5882–5897 (2013)
47. Reingold, O., Rothblum, G.N., Rothblum, R.D.: Constant-round interactive proofs for delegating computation. SIAM J. Comput. **50**(3) (2021). https://doi.org/10.1137/16M1096773
48. Setty, S.T.V.: Spartan: Efficient and general-purpose zkSNARKs without trusted setup. In: Micciancio, D., Ristenpart, T. (eds.) CRYPTO 2020. LNCS, vol. 12172, pp. 704–737. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-56877-1_25
49. Setty, S.T.V., Thaler, J., Wahby, R.S.: Unlocking the lookup singularity with lasso. In: Joye, M., Leander, G. (eds.) EUROCRYPT 2024. LNCS, vol. 14656, pp. 180–209. Springer, Cham (2024). https://doi.org/10.1007/978-3-031-58751-1_7
50. Spielman, D.A.: Linear-time encodable and decodable error-correcting codes. IEEE Trans. Inf. Theory **42**(6), 1723–1731 (1996). https://doi.org/10.1109/18.556668

51. Von Zur Gathen, J., Gerhard, J.: Modern Computer Algebra. Cambridge University Press (2003)
52. Wan, Z.X.: Lectures on Finite Fields and Galois Rings. World Scientific Publishing Company (2003)
53. Wee, H., Wu, D.J.: Succinct vector, polynomial, and functional commitments from lattices. In: Hazay, C., Stam, M. (eds.) EUROCRYPT 2023. LNCS, vol. 14006, pp. 385–416. Springer, Cham (2023). https://doi.org/10.1007/978-3-031-30620-4_13
54. Wei, Y., Zhang, X., Deng, Y.: Transparent SNARKs over Galois rings. In: Jager, T., Pan, J. (eds.) PKC 2025. LNCS, vol. 15674, pp. 418–451. Springer, Cham (2025). https://doi.org/10.1007/978-3-031-91820-9_14
55. Zeilberger, H.: Khatam: reducing the communication complexity of code-based SNARKs. IACR Cryptology ePrint Archive, p. 1843 (2024). https://eprint.iacr.org/2024/1843